



A non-contact material recognition method using ultrasonic echo signals and deep learning^{☆,☆☆}

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ABSTRACT

Accurate wall material recognition is essential for robotic operations in extreme environments such as mining tunnels, disaster sites, and search-and-rescue missions, where conventional sensors like cameras and LiDAR often fail due to darkness, dust, smoke, or obstructions. Ultrasonic sensing offers a robust alternative, but its echo signals exhibit complex spatial-temporal patterns that are difficult to model with traditional methods. This study proposes AE-CS-TCN (Attention-Enhanced Cross-Scale Temporal Convolutional Network), a deep learning architecture for non-contact wall material recognition using raw ultrasonic echoes. The model integrates spatial attention, dilated temporal convolutions, cross-scale fusion, and cross-attention to effectively capture and align multi-resolution features. Experiments on both the public LMT dataset and a self-built dataset show that AE-CS-TCN achieves 96% average accuracy, outperforming conventional and deep learning baselines while maintaining strong robustness to noise and distance variations.

1. Introduction

Reliable material recognition in extreme environments constitutes a fundamental capability for autonomous robots deployed in high-stakes operational scenarios (such as nuclear facility maintenance, subterranean exploration, and post-disaster response missions). In these challenging settings, robots routinely engage in physical interaction with structural elements, including walls, pillars, support frames, and protective enclosures, to carry out critical tasks ranging from structural inspection and object manipulation to safe navigation through confined or degraded spaces [1–6]. Accurate and timely inference of the underlying material properties (such as elasticity, density, surface roughness, and acoustic reflectivity), is essential not only for safeguarding the robot and its surroundings during contact but also for enabling intelligent decision-making in dynamic and uncertain conditions. This material-aware perception directly informs adaptive control policies, contact planning, and actuation strategies tailored to the mechanical characteristics of the encountered surfaces [2,7,8]. By aligning robotic behavior with the physical attributes of the environment, such adaptation enhances task execution fidelity, minimizes mechanical wear, and reduces unnecessary energy expenditure. Crucially, in environments

where visual sensing is compromised (due to poor lighting, occlusion, smoke, dust, or radiation), reliance on multimodal or non-visual sensing modalities becomes imperative. Therefore, robust material perception under adverse sensing conditions is not merely advantageous but indispensable for ensuring the autonomy, safety, and operational efficacy of robots in complex real-world domains.

However, these operational environments are typically characterized by severely degraded sensing conditions that fundamentally challenge conventional perception systems. Factors such as low illumination, dense dust, smoke, and elevated radiation levels substantially compromise the reliability of traditional optical sensing modalities [9–11]. Cameras, in particular, often suffer from low signal-to-noise ratios, motion blur, and strong light scattering, all of which degrade image quality and impede robust visual feature extraction. Likewise, LiDAR systems experience significant signal attenuation, elevated measurement noise, and data sparsity due to beam scattering, occlusion, or absorption by airborne particulates [10,12–15]. These sensing degradations not only diminish the accuracy of geometric reconstruction and semantic understanding but also erode the reliability of downstream autonomy functions, including localization, mapping, and navigation.

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Consequently, the deployment of robotic systems in mission-critical scenarios remains limited by the inherent fragility of optical sensing under such adverse conditions. This limitation highlights the pressing need for alternative or complementary sensing paradigms that possess intrinsic robustness to environmental disturbances and can deliver dependable perceptual feedback precisely in the regimes where vision-based systems falter [1,7,16,17].

Among the various alternative sensing modalities, ultrasonic sensing distinguishes itself through its robustness in visually degraded and electromagnetically noisy environments. It functions reliably under extreme lighting conditions, maintains consistent performance in the presence of airborne particulates such as dust and smoke, and demonstrates minimal susceptibility to electromagnetic interference [18–21]. More importantly, ultrasonic echoes carry rich physical signatures that reflect intrinsic material properties, including surface texture, acoustic impedance, porosity, and internal layering [22–24]. When an ultrasonic wave interacts with a material surface, the reflected signal encodes high-dimensional information spanning spatial, temporal, and spectral domains [9,21,25–29]. Spatial features, derived from beamforming or array-based measurements, reveal surface morphology and geometric discontinuities. Temporal characteristics capture the dynamic evolution of the echo, offering insights into wave damping, material elasticity, and time-of-flight delays. Concurrently, multi-scale frequency analysis enables discrimination across structural hierarchies: high-frequency components are sensitive to fine surface irregularities, whereas low-frequency content conveys information about deeper bulk properties [1,7,17,25,30]. Together, these complementary dimensions constitute a comprehensive and physically grounded representation that underpins robust material recognition in challenging operational environments.

Traditional machine learning approaches to ultrasonic material recognition have largely depended on handcrafted features informed by domain-specific knowledge, such as time-of-flight, peak amplitude, zero-crossing rate, and spectral centroid [7,16,31]. While these methods offer benefits in interpretability and computational efficiency, they suffer from several fundamental limitations. First, they often oversimplify the rich physical complexity embedded in ultrasonic echo signals, discarding subtle yet discriminative patterns that are critical for accurate material identification. Second, their fixed feature representations lack the flexibility to adapt to varying environmental conditions, sensor configurations, or diverse material classes. Third, they do not scale effectively to large or heterogeneous datasets, particularly when confronted with variable-length signals or inconsistent acquisition protocols [7,11,21,24,27,32,33]. Furthermore, conventional classifiers (such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Decision Trees) operate on static feature vectors and are inherently incapable of modeling temporal dynamics or hierarchical interactions among signal components. As a result, they fail to capture the intricate spatiotemporal structure inherent in ultrasonic data, which significantly constrains their discriminative capacity in real-world, unstructured environments [1,7,22,34,35].

In recent years, deep learning has emerged as a powerful paradigm for ultrasonic material recognition by enabling end-to-end learning of discriminative representations directly from raw or minimally processed sensor data. Convolutional Neural Networks (CNNs) have shown promise in capturing localized spatial patterns from time-domain waveforms or time-frequency representations such as spectrograms [8, 30,36]. At the same time, sequential models including Long Short-Term Memory (LSTM) networks and Temporal Convolutional Networks (TCNs) have been employed to model temporal dependencies in echo sequences [1,20,34,37]. Despite these advances, current deep architectures still encounter significant limitations. Standard CNNs possess limited receptive fields and are poorly equipped to model long-range temporal correlations that are essential for characterizing complex material responses. Recurrent models, while theoretically capable of capturing temporal dynamics, often suffer from vanishing gradients,

slow convergence during training, and high computational costs. Furthermore, most existing approaches do not explicitly account for the multi-scale nature of ultrasonic signatures (where fine-grained surface effects and coarse bulk properties manifest at different temporal and spectral resolutions) nor do they incorporate adaptive mechanisms to selectively attend to the most informative segments of the signal [1,7, 20,22,34,35,37]. As a result, the performance of these models remains suboptimal, particularly under the adverse sensing conditions typical of extreme environments.

To address the aforementioned limitations, we propose a novel deep learning architecture named the Attention-Enhanced Cross-Scale Temporal Convolutional Network (AE-CS-TCN) for robust ultrasonic material recognition in extreme environments. AE-CS-TCN is specifically designed to jointly learn and fuse spatial, temporal, and multi-scale features directly from raw ultrasonic echo signals, exploiting the intrinsic physical structure of acoustic reflections. The architecture integrates four synergistic components: (1) a spatial attention block that adaptively emphasizes discriminative spatial regions; (2) a temporal convolutional block that efficiently captures short- and long-range temporal dynamics; (3) a cross-scale fusion block that aggregates complementary information across multiple temporal resolutions; and (4) a cross-attention block that models interdependencies among signal segments to enhance discriminative focus and contextual awareness. To validate the proposed architecture, we developed a comprehensive experimental platform centered on a robotic system equipped with custom ultrasonic sensing modules. We collected a new dataset of ultrasonic echoes spanning multiple material categories under controlled and realistic conditions. Extensive experiments were conducted to evaluate AE-CS-TCN against other approaches, using both the publicly available LMT dataset and the dataset introduced in this work, across diverse material types and environmental scenarios.

The main contributions of this work are summarized as follows:

- We propose AE-CS-TCN, a new architecture that integrates spatial attention, temporal modeling, cross-scale feature fusion, and cross-attention mechanisms for effective ultrasonic material recognition in extreme conditions.
- We construct a custom experimental system and dataset using a robot platform, enabling reproducible evaluation of ultrasonic-based material recognition in realistic environments.
- We conduct extensive comparative experiments demonstrating that AE-CS-TCN consistently outperforms existing methods in terms of recognition accuracy, robustness to noise and occlusion, and generalization across material types.

The remainder of this paper is organized as follows. Section 2 reviews related work in ultrasonic sensing and learning-based material recognition. Section 3 introduces the AE-CS-TCN architecture and its constituent blocks. Section 4 describes the dataset and experimental setup. Section 5 presents the results and performance evaluations. Section 6 concludes the paper and outlines directions for future research.

2. Related work

Material classification using ultrasonic echo signals remains a challenging yet highly impactful research direction, primarily due to material heterogeneities, surface irregularities, and overlapping reflections that obscure discriminative signal patterns. Over the years, a range of methodological approaches has been developed to tackle these challenges. Early efforts predominantly employed traditional machine learning pipelines, which first extract handcrafted features and subsequently feed them into classifiers like support vector machines or random forests [7,20,21,34]. These methods benefit from strong interpretability and often achieve competitive performance even with limited training data. More recently, deep learning has emerged as a

dominant paradigm by enabling end-to-end learning directly from raw waveforms or their time–frequency representations, such as spectrograms. Architectures based on CNNs and recurrent models (including LSTMs and temporal convolutional networks), automatically discover hierarchical, task-specific features that exhibit greater robustness to noise, environmental variability, and sensor imperfections [1,8,17,30,36–38]. In this section, we provide a structured overview of these two key strategies: (1) traditional machine learning approaches grounded in domain-informed, handcrafted morphological and statistical features, and (2) modern deep learning techniques that perform automatic, end-to-end representation learning.

2.1. Traditional machine learning

Traditional machine learning techniques play a pivotal role in the classification of materials based on ultrasonic signal analysis, forming a foundational framework in the domain of ultrasonic-based material recognition [7,20,21,32,34,39]. These approaches generally involve two key stages: feature extraction and model-based classification. In the feature extraction phase, meaningful characteristics are manually engineered from raw ultrasonic signals using domain-specific knowledge and signal processing techniques. Time-domain features such as peak amplitude, echo duration, and standard deviation have been widely adopted, particularly in studies involving pavement surface classification using backscattered ultrasound signals. For example, Bystrov et al. [20,21] develop a material identification method using backscattered ultrasound and radar signals, enabling the classification of pavement surface materials based on differences in the backscattered characteristics, such as peak value, duration, and standard deviation of the echo signals. This approach demonstrates the effectiveness of time-domain features in distinguishing materials with distinct acoustic reflectivity. In parallel, frequency-domain analysis has been extensively explored, leveraging the unique frequency response patterns exhibited by different materials. Researchers such as Hideto [32] and Smith [31] show that different materials exhibit distinct frequency response patterns, which can be effectively used for material classification. For instance, Hideto [32] applies frequency-domain features extracted from ultrasonic echoes to classify industrial materials, achieving high accuracy by exploiting the spectral differences between metallic and non-metallic surfaces. Similarly, Smith [31] uses frequency-domain analysis to distinguish construction materials such as concrete, wood, and steel, based on their resonant frequency characteristics.

In the classification stage, a variety of traditional machine learning models are employed to map the extracted features to specific material classes. Among the most commonly used are KNN, SVM, Decision Trees, and so on. The KNN algorithm has been widely applied due to its simplicity and ease of implementation, particularly in small-scale classification tasks involving ultrasonic features derived from time-domain analysis. For instance, Smith [31] applies KNN to classify construction materials using ultrasonic features derived from time-domain analysis. Although this approach shows promising results in controlled environments, its performance often degrades in more complex and variable settings due to the inherent noise and variability of ultrasonic signal characteristics [7,20,21]. In contrast, SVM demonstrates superior accuracy and robustness, especially in handling high-dimensional feature spaces and noisy data. The work of Geng [7], for instance, shows that SVM can effectively classify a range of materials using frequency-domain features, although challenges remain in distinguishing materials with similar acoustic properties, leading to occasional misclassification errors. Decision Trees and Random Forests, on the other hand, offer the advantage of modeling non-linear relationships and handling complex decision boundaries. Studies by Tiitta [39] and Nasir [34] highlight the effectiveness of Random Forests in classifying materials with intricate surface textures using wavelet-transformed ultrasonic features. For example, Nasir [34] employs a Random Forest classifier to distinguish between different types of composite materials,

achieving high classification accuracy by leveraging wavelet packet transform to extract time–frequency features.

A number of persistent limitations are associated with traditional machine learning methods in the context of ultrasonic-based material recognition. One of the most significant is the reliance on manually engineered features, which makes the performance of these models highly dependent on the quality, relevance, and representativeness of the selected features. This dependency often results in poor generalization when models are applied to new materials or in environments with varying acoustic conditions [11,20,21,34,37,39]. Additionally, traditional approaches struggle to capture the inherently non-linear and multi-scale characteristics of ultrasonic signals, which are often rich in temporal and spectral information. As a result, these methods may underperform in complex or noisy environments where subtle variations in material properties are difficult to distinguish using hand-crafted features. Moreover, the requirement for extensive preprocessing and manual feature selection increases the overall complexity of the pipeline and limits the scalability and real-time applicability of these techniques [11,40]. Deep learning techniques, which are capable of automatically learning hierarchical and discriminative features directly from raw ultrasonic signals, offer a promising alternative [32,35,41–43].

2.2. Deep learning

Recent advances in deep learning have opened new avenues for ultrasonic-based material identification by enabling end-to-end feature extraction directly from raw or transformed signals [1,8,17,30,36–38,44,45]. For example, Eisele [44] combines high-precision ultrasonic phase measurement with a pulse neural network architecture to build a system that can reliably extract and classify target information from echo data. Moreover, recent studies extend this framework to material identification. In another study, Kang [1] proposes a novel method for material property recognition by mimicking the acoustic recognition mechanisms of animals like bats and dolphins, which rely on returning echoes of self-emitted acoustic signals. The approach employs an end-to-end machine learning framework using Convolutional Neural Networks to convert spectral cues from reflected echoes into meaningful information about target objects. Two experiments are conducted for validation: the first aims to learn predefined everyday surfaces using reflected acoustic signals emitted from a distance, and the second extends this to classify acoustically smooth surfaces (e.g., liquid solutions). Results show the system can recognize texture and density information, with the CNN-based approach outperforming conventional feature-engineering-dependent machine learning methods by achieving robust high accuracies (> 90%) on test datasets. Additionally, Sahoo [36] explores the potential of non-contact ultrasonic sensors beyond range measurement and object detection, highlighting that material type identification is vital for robotic navigation and autonomous vehicle applications. Ultrasonic echo signals, which contain rich information, are used to categorize materials such as glass, wood, metal plates, sponges, and cloth. The method involves extracting envelopes from raw echo signals via Hilbert transform and employing a novel one-dimensional Convolutional Neural Network (1D-CNN) architecture for classification. Evaluated using metrics like accuracy, precision, recall, and F1-score, the proposed 1D-CNN model achieves impressive performance with 96% accuracy, 95% precision, 95% recall, and 95% F1-score, demonstrating the effectiveness of deep learning in material classification using ultrasonic signals. Researchers [8] successfully apply deep learning models to ultrasonic echo signals for precise classification of metal materials. By using other deep learning architectures, their models achieve robust performance in distinguishing between various materials.

2.3. Limitations of existing research

Despite significant advances in echo signal analysis and processing, current methodologies continue to exhibit critical limitations that constrain their effectiveness in real-world robotic perception tasks. One major shortcoming is their inability to extract meaningful spatial information from echo returns. Spatial context is essential for reconstructing the geometric configuration of a scene, resolving the relative positions of multiple targets, and inferring structural relationships among objects. Without explicit spatial modeling, systems struggle to support tasks such as localization, mapping, and obstacle avoidance in complex environments.

Equally important is the inadequate treatment of temporal dynamics in existing approaches. Echo signals from moving targets inherently encode time-dependent characteristics, including velocity, acceleration, and behavioral patterns. However, many conventional methods either process echoes independently across time or apply simplistic temporal filtering that fails to capture the underlying motion dynamics. This neglect of temporal structure limits the system's ability to track targets reliably or predict their future states, especially in cluttered or dynamic settings.

A further challenge arises from the multi-scale nature of echo signals across different sensing ranges. Signals reflected from nearby, mid-range, and distant targets exhibit substantial differences in amplitude, bandwidth, time-of-flight resolution, and scattering behavior. Yet, most existing techniques rely on fixed or scale-agnostic processing pipelines that cannot adapt to these variations. As a consequence, critical details may be lost at certain ranges, leading to incomplete or inconsistent environmental representations that compromise downstream decision-making.

In addition, prevailing methods generally lack the capacity to identify and emphasize task-relevant signal components. Rather than selectively attending to informative features such as transient scattering events, characteristic frequency modulations, or Doppler signatures that carry discriminative target information, they often treat all signal segments uniformly. This indiscriminate processing not only obscures salient cues but also amplifies the impact of noise and irrelevant clutter, thereby reducing the overall robustness and accuracy of signal interpretation.

Taken together, these limitations result in signal representations that are shallow, fragmented, and insufficient for capturing the rich spatiotemporal structure of real-world echo data. The absence of integrated spatial reasoning, dynamic temporal modeling, adaptive scale handling, and feature-aware processing severely restricts the applicability of current systems in practical robotic scenarios, where environmental complexity, target diversity, and operational uncertainty are the norm.

To overcome these challenges, it is imperative to develop advanced deep learning frameworks specifically designed for echo signal understanding. Such frameworks should incorporate mechanisms for hierarchical spatial encoding, temporal coherence modeling, multi-scale feature fusion, and learnable attention to salient signal components. By jointly optimizing these capabilities within a unified architecture, it becomes possible to achieve more comprehensive, robust, and semantically grounded interpretations of echo signals. This, in turn, will enable robotic systems to operate more effectively in diverse, unstructured, and dynamic environments.

3. Methodology

3.1. Rationale for the AE-CS-TCN architecture

The limitations discussed in the previous section reveal four critical gaps in current approaches to ultrasonic echo analysis: an inability to extract spatial structure (G1), a lack of effective modeling for long-range temporal dynamics (G2), insufficient handling of multi-scale

signal characteristics (G3), and a failure to focus on the most discriminative parts of the signal (G4). To address these challenges in a unified and principled manner, we propose the attention enhanced cross scale temporal convolutional network, a domain-specialized architecture designed specifically for robust, non-contact material recognition from raw ultrasonic echo waveforms.

To explicitly address these four critical gaps (G1–G4) and ensure each design choice is grounded in resolving a specific limitation of prior work, we establish a one-to-one mapping between research gaps and the architectural blocks of AE-CS-TCN. This mapping not only clarifies the causal relationship between problem identification and solution design but also provides clear traceability to subsequent sections detailing block implementations. Table 1 summarizes this gap-block correspondence, with cross-references to subsections that elaborate on each block's structure and working mechanism.

- **G1 (Inability to Extract Spatial Structure):** Ultrasonic echoes, though inherently one-dimensional in the time domain, implicitly encode spatial information through physical phenomena such as time-of-flight differences, beam divergence, and multipath reflections. This spatial context, critical for distinguishing materials with similar temporal envelopes but distinct surface morphologies, remains unexploited by existing methods. This gap is closed by the spatial attention block, which learns to emphasize geometrically informative regions within feature representations, endowing the model with spatial awareness derived directly from raw waveforms.
- **G2 (Lack of Effective Long-Range Temporal Dynamics Modeling):** Material-specific signatures in ultrasonic echoes extend beyond initial reflection peaks to include reverberation tails and internal resonances, which may span thousands of signal samples. Capturing these patterns requires a model that maintains temporal causality, avoids the sequential bottlenecks of recurrent networks, and supports a large receptive field. This gap is addressed by the temporal convolutional block, which leverages dilated causal convolutions to enable efficient, parallelized learning of both transient and sustained temporal dynamics over extended time horizons.
- **G3 (Insufficient Handling of Multi-Scale Signal Characteristics):** Ultrasonic echoes exhibit inherent multi-scale properties: high-frequency modulations reflect fine surface textures (e.g., fabric weave, paper fibers), while low-frequency energy distributions correlate with bulk material attributes (e.g., acoustic impedance, internal damping). In real-world robotic scenarios, variable sensor-to-target distances cause the dominant informative scale to shift dynamically, which fixed-receptive-field models cannot adapt to. This gap is resolved by the cross-scale block, which processes signals through parallel convolutional branches with hierarchically increasing dilation rates, enabling simultaneous capture of features at multiple temporal granularities.
- **G4 (Failure to Focus on Discriminative Signal Parts):** Not all segments of an ultrasonic echo carry equal material-relevant information—primary reflection peaks and resonant frequencies are typically most informative, while late reverberation or background noise can degrade performance. Existing methods treat all time steps uniformly, obscuring salient cues and amplifying noise. This gap is mitigated by dual attention mechanisms: the cross-attention block adaptively amplifies informative time segments while suppressing noise-dominated intervals, together enhancing model robustness without increasing complexity.

Importantly, the entire AE-CS-TCN architecture operates end-to-end on raw time-domain signals. By avoiding handcrafted features (e.g., spectrograms, Mel Frequency Cepstrum Coefficients), the model preserves phase coherence and microsecond-level transient details critical for differentiating acoustically similar materials. This design is

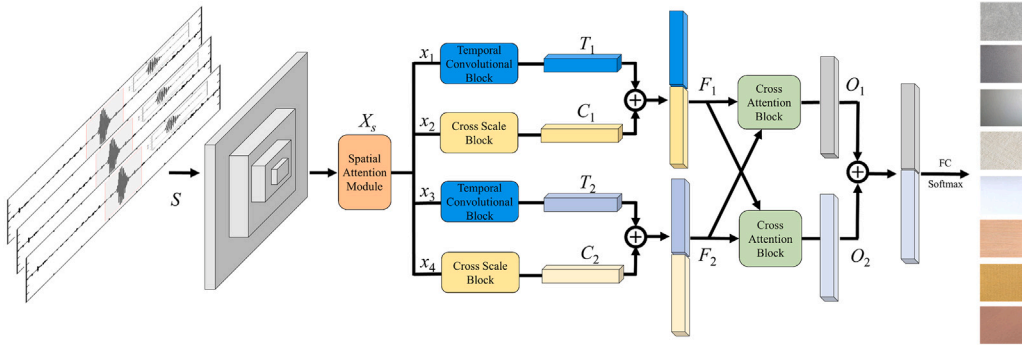


Fig. 1. The overall frame of the AE-CS-TCN model.

Table 1

Mapping between recognition challenges in ultrasonic echoes and AE-CS-TCN modules.

Gap ID	Identified limitation	Proposed module	Section
G1	An inability to extract spatial structure	Spatial Attention Block	Section 3.3
G2	A lack of effective modeling for long-range temporal dynamics	Temporal Convolutional Block	Section 3.4
G3	Insufficient handling of multi-scale signal characteristics	Cross Scale Block	Section 3.5
G4	A failure to focus on the most discriminative parts of the signal	Cross attention Block	Section 3.6

Table 2

Detailed network structure setting of AE-CS-TCN.

Network layer	Branch/Process	Input size (Channels $\times$ Length)	Output size (Channels $\times$ Length)
Raw Input	Main Process	–	1 $\times$ 1400
Convolution	Main Process	1 $\times$ 1400	64 $\times$ 1400
Spatial Attention Block	Main Process	64 $\times$ 1400	64 $\times$ 1400
Temporal Convolutional Block	Branch 1	64 $\times$ 1400	512 $\times$ 1
Cross Scale Block	Branch 2	64 $\times$ 1400	512 $\times$ 8
Concatenation	Fusion Layer	512 $\times$ 1 + 512 $\times$ 8	1024 $\times$ 8
Temporal Convolutional Block	Branch 3	64 $\times$ 1400	512 $\times$ 1
Cross Scale Block	Branch 4	64 $\times$ 1400	1024 $\times$ 8
Concatenation	Fusion Layer	512 $\times$ 1 + 512 $\times$ 8	1024 $\times$ 8
Cross Attention Block	Fusion Layer	1024 $\times$ 8 + 1024 $\times$ 8	1024 $\times$ 8 + 1024 $\times$ 8
Concatenation	Fusion Layer	1024 $\times$ 8 + 1024 $\times$ 8	2048 $\times$ 8
Convolution	Output Layer	2048 $\times$ 8	1 $\times$ 8
Final Output	Output Layer	1 $\times$ 8	1 $\times$ 8

not a generic sequence model repurposed for ultrasonic sensing but a physics-informed framework, where each component is aligned with the physical principles of ultrasonic wave–material interaction and targeted to resolve a specific identified gap.

3.2. The AE-CS-TCN model

Fig. 1 depicts the overall architecture of the proposed deep model, which comprises four interlinked blocks: the spatial attention block, the temporal convolution block, the cross scale block, and the cross attention block. Initially, the raw echo waveform S is passed through a shallow convolutional layer to extract low-level feature maps that retain fine-grained signal characteristics. These preliminary features are then routed into the spatial attention block, where a learnable spatial mask is computed to emphasize geometrically salient regions of the echo data, yielding an attention-weighted feature map X_s .

Subsequently, X_s is uniformly partitioned into four sub-tensors $\{x_1, x_2, x_3, x_4\}$ along the channel dimension. The first and third sub-tensors (x_1, x_3) are processed by the temporal convolution block with distinct dilation rates and kernel sizes to capture both short- and long-range dynamic patterns, producing temporal embeddings T_1 and T_2 . In parallel, the second and fourth sub-tensors (x_2, x_4) enter the cross scale block, which applies multi-branch convolutions at varying receptive fields to extract frequency-dependent features C_1 and C_2 . The outputs T_1 and C_1 are concatenated to form the intermediate vector F_1 , while T_2 and C_2 are fused into F_2 .

A cross attention block is then employed to model high-order semantic correlations between F_1 and F_2 . This mechanism computes pairwise affinity scores and generates context-aware outputs O_1 and O_2 , effectively integrating temporal, spatial, and scale information. Finally, the concatenated vector is fed into a fully connected layer followed by a softmax activation to yield the probability distribution over target classes.

Each functional block contributes a specialized capability: the spatial attention block highlights key structural cues; the temporal convolution block encodes dynamic temporal dependencies; the cross scale block captures multi-frequency phenomena inherent in ultrasonic echoes; and the cross attention block fuses heterogeneous representations to strengthen material-semantic associations. Detailed mathematical formulations and implementation specifics for each block are presented in the following sections.

3.3. Spatial attention block

Spatial attention precisely identifies the most informative spatial regions in an echo signal, thereby enabling the model to extract detailed structural cues such as material composition, morphological characteristics, and boundary delineations [46,47]. In our spatial attention block, the input feature map $s \in \mathbb{R}^{C \times H \times W}$ is first summarized along the channel dimension by two parallel pooling operations (average pooling $\text{AvgPool}(s)$ and max pooling $\text{MaxPool}(s)$) to produce two spatial descriptors. These descriptors capture complementary aspects of the signal: the average pooling emphasizes overall intensity patterns, while

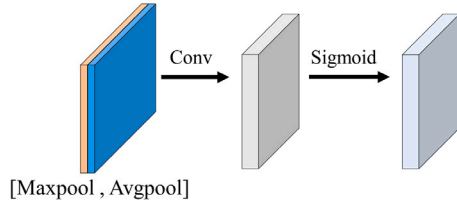


Fig. 2. Spatial attention block.

the max pooling highlights the strongest activations. The two resulting maps are concatenated to form a joint descriptor, which is then convolved with a 7×7 kernel to aggregate local spatial context. Finally, a sigmoid activation normalizes the output into the spatial attention weight map $X_s \in (0, 1)^{1 \times H \times W}$, which reweights the original feature map and enhances representation of salient regions. Fig. 2 illustrates this workflow. By integrating both broad and focused pooling cues before convolution, the block balances global structure awareness with local detail sensitivity. This design not only strengthens the ability of the network to capture intricate spatial details in ultrasonic echoes but also stabilizes training by avoiding excessive parameter growth in the attention mechanism. The equation for computing spatial attention is as follows:

$$m = \text{concat}(\text{Avgpool}(s); \text{Maxpool}(s)) \quad (1)$$

$$X_s(s) = \text{sigmoid}(K^{7 \times 7}(m)) \quad (2)$$

where $K^{7 \times 7}$ denotes a convolution operation with a convolution kernel size of 7×7 .

3.4. Temporal convolutional block

Temporal convolutional blocks are uniquely suited for ultrasonic echo analysis, as they excel at modeling both short-term and long-term temporal dependencies inherent in such signals. Ultrasonic echoes exhibit complex time-domain characteristics: short-term dynamics include transient pulses from material interfaces, while long-term patterns involve gradual amplitude decay due to energy dissipation in the medium. Temporal convolutional blocks address these dual requirements efficiently, avoiding the computational overhead of recurrent networks while maintaining comparable modeling power. Fig. 3 shows the details of the temporal convolution block, comprising a dilated causal convolution layer, residual connection, normalization layer, dropout, and an activation function. This block has two key operations: the dilated causal convolution and the residual connection. This design enables deep feature extraction from temporal sequences while ensuring stable training dynamics, which is critical for preventing performance degradation in deep architectures.

3.4.1. Dilated causal convolution

The core of the temporal convolutional block lies in its dilated causal convolution layer, whose structure is detailed in Fig. 3. This layer synergistically combines causal convolution and dilated convolution to balance temporal locality and long-range dependency modeling. Causal convolution enforces a strict time-constrained processing order, where each output at time t depends solely on inputs from times $t' \leq t$. This unidirectional flow aligns with the physical nature of ultrasonic data acquisition, where future signal samples (relative to the current time step) are unavailable during real-time analysis, thus preserving temporal causality.

Dilated convolution enhances this framework by introducing skip sampling between input elements, controlled by a dilation factor d . For a given sequence position i , the input x is sampled at intervals of d , effectively expanding the receptive field without increasing filter size.

This design is particularly advantageous for ultrasonic signals: a small dilation factor ($d = 2$) captures fine-grained short-term dynamics, while larger factors ($d = 8$) extend the receptive field to capture long-term trends. The mathematical formulation of the dilated causal convolution operator is provided in Eq. (3):

$$F(t) = \sum_{i=0}^{k-1} f(i) \cdot x_{t-d \cdot i} \quad (3)$$

where k denotes the filter size, d denotes the dilated factor, x denotes the input sequence, $F(t)$ denotes the output of the dilated causal convolution at t , and $f(i)$ denotes the weight of the filter at position i .

3.4.2. Residual connections

In deep networks processing long temporal sequences, gradient instability (manifested as vanishing or exploding gradients) can severely degrade training efficiency. Residual connections mitigate this issue by introducing a shortcut path that directly propagates the input signal to deeper layers, stabilizing gradient flow across network depth. Within the temporal convolutional block, the residual connection operates as follows: the input feature map x is combined with the transformed feature map generated by the sequence of dilated causal convolutions, layer normalization, nonlinear activation (e.g., GELU), and dropout. This combination forms a residual function that learns incremental feature refinements rather than reconstructing the entire feature space from scratch. Mathematically, the residual output is defined as:

$$T(x) = F(x) + x \quad (4)$$

where $F(x)$ represents the residual function and x represents the input data.

3.5. Cross scale block

The ultrasonic echo data inherently encapsulates a rich spectrum of echo information spanning diverse frequencies and time periods, where high-frequency components often correspond to fine-grained material textures and low-frequency components reflect macroscopic structural properties [7,19,22]. Consequently, the extraction of signatures across these multi-scale characteristics is paramount for achieving robust and accurate material recognition of the target object. To address this critical requirement, our model incorporates a cross scale block, as illustrated in Fig. 4, specifically engineered to capture and integrate multi-scale information from the ultrasonic echo signal.

The cross scale block is designed to not only decompose and model the echo data but also adaptively weight the significance of features at different scales, thereby enhancing the discriminative power of extracted representations. This adaptability is particularly critical in scenarios where the transducer-target distance varies dynamically: when the transducer is in close proximity, the echo signal exhibits high spatial resolution within a confined field of view, enabling the capture of subtle material inhomogeneities (corresponding to small-scale features) but with potential susceptibility to noise due to near-field interference; conversely, at greater distances, the signal covers a broader field but suffers from reduced resolution and increased attenuation, necessitating the extraction of large-scale structural features to maintain recognition accuracy. The cross scale block effectively mitigates these distance-induced trade-offs by dynamically balancing the extraction of fine and coarse details, ensuring consistent performance across varying operational conditions.

The cross scale block is structured around a hierarchical, residual-like connectivity pattern that connects distinct filter groups. This architecture enables the effective modeling of receptive fields across a range of scale combinations, reflecting the multi-scale nature of human visual perception. The block operates as follows: initially, the input feature map undergoes a 1×1 convolution to perform channel transformation, a step that reduces computational complexity by projecting the feature

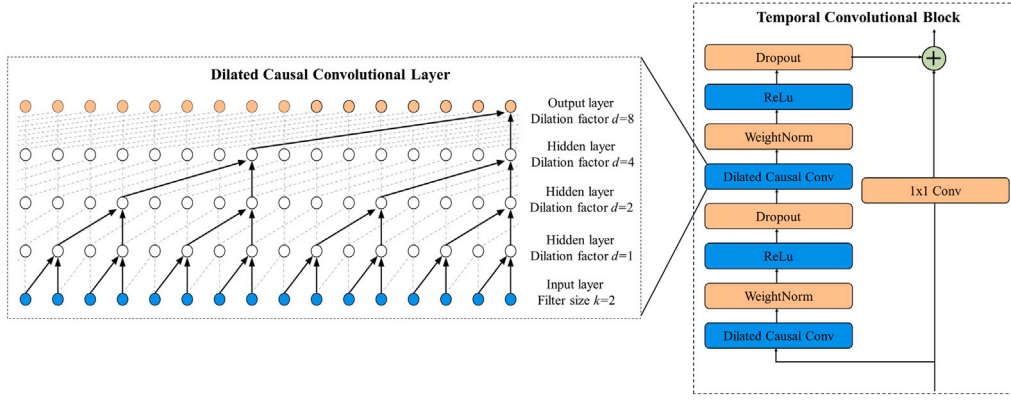


Fig. 3. Temporal convolutional block as well as the dilated causal convolution.

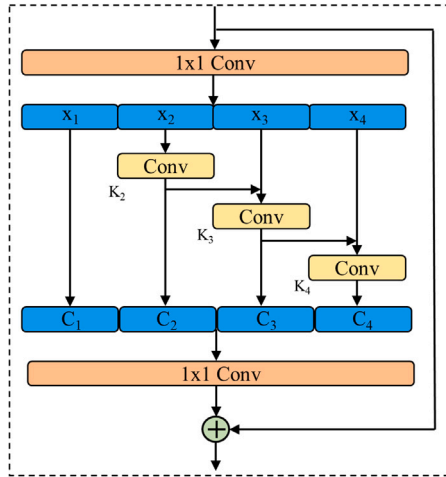


Fig. 4. Cross scale block.

space into a more compact dimension while preserving critical semantic information. The transformed feature map is then uniformly partitioned into s subsets, denoted as x_i where $i \in \{1, 2, \dots, s\}$. Each subset x_i retains the same spatial dimensions as the input but contains $1/s$ of the original channels, with s serving as a scale control parameter (typically set to $2 \leq s \leq 8$ based on empirical analysis of ultrasonic signal bandwidth).

With the exception of the first subset x_1 , each subsequent subset x_i undergoes a 3×3 convolution (chosen for its efficiency in balancing receptive field expansion and computational cost compared to larger kernels like 5×5) and is sequentially integrated with the output of the preceding subset. This sequential aggregation mechanism enables the progressive expansion of receptive fields, capturing increasingly larger-scale features. Finally, the processed multi-scale features are fused with the original input feature map via a 1×1 convolution, which reprojects the concatenated channels into the target dimension and reinforces residual connections to alleviate gradient vanishing during training. Mathematically, the output C_i of the cross scale block is defined as:

$$C_i = \begin{cases} x_i & i = 1 \\ K_i(x_i) & i = 2 \\ K_i(x_i + C_{i-1}) & 2 < i \leq s \end{cases} \quad (5)$$

where $K_i()$ is a 3×3 convolution corresponding to x_i . When $i > 2$, the input of $K_i()$ contains the output of the feature subset x_i and $K_{i-1}()$, a residual connection that facilitates the propagation of both low-level (fine-scale) and high-level (coarse-scale) information throughout the block.

Notably, this hierarchical aggregation induces a combinatorial expansion of receptive fields: each convolution $K_i(\cdot)$ implicitly incorporates information from all preceding subsets, enabling the modeling of scale combinations such as $x_1 + x_2$, $x_1 + x_2 + x_3$, etc. As each x_i undergoes 3×3 convolution, its receptive field expands by a factor proportional to i , leading to a diverse set of receptive field sizes that match the multi-scale nature of ultrasonic echoes. Through this multi-branch integration, the cross scale block achieves effective multi-scale feature fusion, significantly enhancing the model's ability to characterize materials across varying levels of detail and ultimately improving recognition accuracy [48–51].

3.6. Cross attention block

Notably, neither the temporal convolution block nor the cross scale block includes mechanisms to emphasize critical components in their outputs. This limitation can hinder the extraction of discriminative features that are essential for material recognition. Specifically, the temporal convolution block is designed to model sequential dependencies in ultrasonic echo signals but does not prioritize time-domain signatures that are strongly correlated with material properties. Similarly, the cross scale block captures multi scale spatial features but does not highlight the scale specific patterns that are most informative for recognition tasks. Furthermore, a critical issue arises in the integration of the outputs from these two blocks. The temporal features T_i , which encode time varying echo dynamics, and the cross scale features C_i , which encode multi resolution spatial structures, remain processed independently. As a result, the model is unable to learn the interdependencies between temporal evolution and spatial characteristics, which are often crucial for achieving accurate material classification.

The attention mechanism addresses these limitations by adaptively weighting relevant input features, focusing on critical information while suppressing noise. This is particularly valuable in the analysis of ultrasonic data, where material signatures are embedded within complex echo patterns. As illustrated in Fig. 5(a), input vectors x are projected through learnable parameters W_k , W_q , and W_v to generate the query Q , key K , and value V . The similarity between Q and K^T , computed using a dot product or scaled dot product, produces the attention weights W . These weights are then applied to V via matrix multiplication to yield the attention output. The scaled dot product formulation is employed to mitigate gradient instability in high-dimensional spaces. While standard attention computes intra sequence correlations, cross attention models inter sequence dependencies, making it particularly suitable for fusing T_i and C_i . Material properties often manifest through coupled temporal spatial patterns, and cross attention effectively captures such relationships, as shown in Fig. 5(b). In our cross attention block, the Q and K^T from the F_1 vector are first used to compute the attention weight W . Then, W is dot-multiplied with the V from the

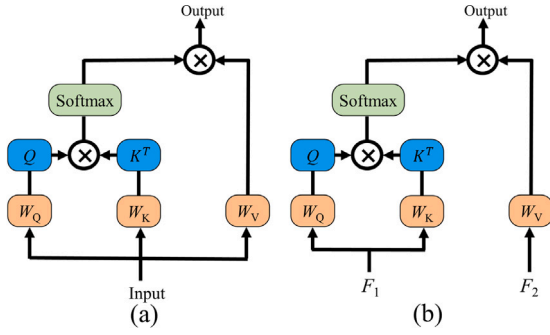


Fig. 5. Different attention mechanisms. (a) is the self attention, (b) is the cross attention.

F_2 vector to obtain a cross attention score, enabling the model to learn interdependencies between temporal and spatial features. The formula for the attention weighting coefficient is as follows:

$$\begin{cases} Q = x \cdot W_q \\ K = x \cdot W_k \\ V = x \cdot W_v \end{cases} \quad (6)$$

$$W = \text{softmax}(QK^T) \quad (7)$$

$$\text{Attention}(Q, K, V) = W \cdot V = \text{softmax}(QK^T) \cdot V \quad (8)$$

4. Dataset

4.1. LMT dataset

In this study, we utilize the LMT dataset introduced by Strese et al. in 2017 and 2019 [52,53], which captures synchronized multi modal signals during controlled interactions between tools and surfaces. These signals include vibrations (measured via accelerometer), audio (tapping or dragging sounds), friction forces, and high-resolution flash/no-flash images. The dataset encompasses 108 texture samples, with these surface materials categorized into nine distinct groups: meshes, stone, glossy, wood, rubbers, fibers, foams, foils and papers, textiles and fabrics. For our task, we focus specifically on tapping sound data. Representative examples of each modality across all material classes are illustrated in Fig. 6.

4.2. The dataset constructed in this study

4.2.1. Data collection

In this study, we implement a practical ultrasonic sensing system to acquire echo signals from various materials, with the primary goal of enabling material recognition on a mobile robotic platform. The system is built upon established ultrasonic sensing principles and shares conceptual similarities with prior bio-inspired echo acoustic approaches, such as Kang and Kim [1]. However, while Kang and Kim [1] relied on fixed speaker-microphone arrays in a static laboratory setup with offline signal processing, our work emphasizes the full integration of the sensing system onto the robot itself. As illustrated in Fig. 7, all core components are mounted directly on the mobile platform, enabling self-contained, real-time operation without external instrumentation. This on-robot integration represents the central contribution of our approach.

The core components of the system include: a transmitting ultrasonic transducer (T) to focus energy on the target material; a receiving ultrasonic transducer (R) to capture weak echo signals; a high-frequency ultrasonic control STM32 board responsible for generating

precise 200 kHz excitation pulses and synchronizing transmission/reception timings; an industrial personal computer (IPC) equipped with an 8-channel 16-bit analog-to-digital converter (ADC) operating at 2 MHz sampling rate for high-fidelity data acquisition and processing; a mobile robot with differential drive locomotion (max speed 0.5 m/s) and LiDAR-based positioning (± 1 cm accuracy) for stable movement and pose control; and test materials placed in front of the robot. All core components (transducers, STM32 board, and IPC) are mounted on the frame attached to the mobile robot, minimizing mechanical noise interference and ensuring self-contained signal acquisition in real-time field conditions. The transmitting transducer (T) operates within a frequency range of 200 kHz (± 10 kHz), and exhibits a measurement blind zone in the near field ranging from 0 cm to 25 cm due to acoustic wave interference effects in the transducer's near-field region.

Fig. 8 shows the materials selected for recognition experiments. A total of eight commonly encountered material types (namely, paper, wood, foam, iron, copper, aluminum, stone, and cloth) are chosen to represent diverse acoustic characteristics found in construction, outdoor, and emergency response environments. These materials span a wide range of acoustic impedances and surface roughness (from 0.02 μm for aluminum to 50 μm for stone), resulting in distinct echo patterns: metallic materials (iron, copper, aluminum) produce high-amplitude, low-dispersion echoes due to high reflectivity, while porous materials (paper, foam, cloth) generate low-amplitude, multi-path echoes with significant energy attenuation. Each material sample has uniform physical dimensions (20 cm $\times$ 20 cm in area and 1 mm in thickness) with a flat surface finish (machined to ± 0.1 mm flatness) to ensure consistency and comparability across experimental trials.

The experimental procedure consists of the following main steps. First, the IPC issues motion control commands via a operating system interface, directing the mobile robot to approach and stop at a designated position relative to the test material with a positioning error $\leq \pm 2$ cm. Upon reaching the target position, the STM32 control board triggers the transmitting transducer (T) to emit a broadband ultrasonic impulse signal at 200 kHz, with each pulse having a 0.15-millisecond duration to ensure sufficient signal-to-noise ratio (SNR) at maximum distances. The reflected echo signal is captured by the receiving transducer (R), preprocessed through a 40 dB gain low-noise amplifier, and forwarded to the IPC for storage and subsequent analysis. To construct a balanced and diverse dataset, 200 sets of ultrasonic echo signals are collected for each material type, with each set containing 1400 sampling points to capture reflection signal. This procedure repeats at five different distances from the material surface (30 cm, 35 cm, 40 cm, 45 cm, and 50 cm), covering the near-field to far-field transition region (beyond the 25 cm blind zone) where echo characteristics vary significantly with distance. Environmental conditions are monitored throughout data collection to minimize variations. These data serve as the foundation for subsequent feature extraction, model training, and performance evaluation in ultrasonic-based material classification tasks.

Fig. 9 illustrates the time domain waveforms of ultrasonic echo signals acquired from different materials. As shown in the figure, the signal amplitudes generally fall within the range of 2.3 V to 2.5 V, exhibiting only minor fluctuations across materials. These amplitude differences are marginal and fail to reveal distinct waveform patterns, thereby making it difficult to distinguish material types based solely on visual inspection of the signal shape. Furthermore, the reflected signal intensity shows minimal variation, which limits the capacity of time-domain features to characterize the spatial structure of the materials, such as surface texture or interface continuity.

From a temporal perspective, the echo signals display similar time delays across all tested materials. This uniformity suggests that the time-of-flight (ToF) component of the signal is not sufficiently sensitive to detect subtle variations in acoustic impedance or transmission delay introduced by different material properties. As a result, time-domain

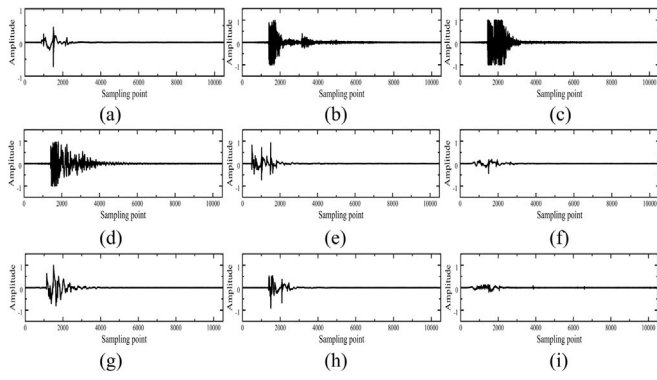


Fig. 6. Tapping Data from LMT dataset. (a) is Meshes, (b) is Stones, (c) is Blank glossy surfaces, (d) is Wood Types, (e) is Rubbers, (f) is Fibers, (g) is Foams, (h) is Foils and Papers, (i) is Textiles and Fabrics.

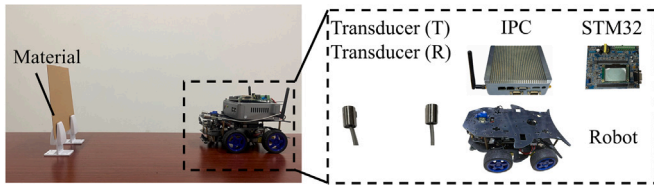


Fig. 7. The equipment of the experimental system.

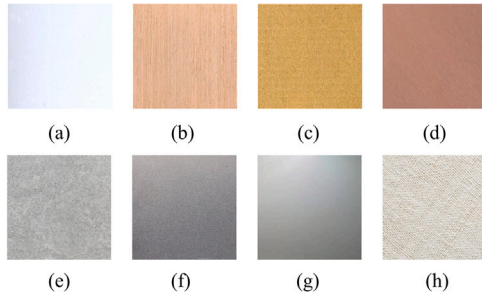


Fig. 8. The materials included in the dataset constructed in this study. (a) is foam, (b) is wood, (c) is paper, (d) is copper, (e) is stone, (f) is iron, (g) is aluminum, (h) is cloth.

features (such as amplitude, pulse width, and delay) offer limited discriminative power when attempting to classify or recognize materials with similar geometric and physical configurations.

Fig. 10 presents the frequency-domain spectra of the ultrasonic echo signals corresponding to the same set of materials. It can be observed that most materials exhibit highly similar spectral profiles, characterized by prominent primary peaks in the 130 kHz to 160 kHz band and secondary peaks in the 200 kHz to 230 kHz band. For the majority of materials, including metals such as copper, iron, and aluminum, the locations and magnitudes of both the primary and secondary peaks are nearly identical. This spectral similarity can be attributed to the uniform size and geometry of the samples, which impose similar boundary reflection and resonance conditions.

However, for materials with high porosity and lower density (such as foam, paper, and cloth), a noticeable reduction in the amplitude of the primary peak is observed. This attenuation is likely due to increased ultrasonic scattering, internal absorption, and energy dispersion as waves propagate through porous or fibrous media. These effects weaken the coherent reflection necessary to maintain high spectral intensity. Nevertheless, even with these subtle differences, the frequency-domain features still demonstrate limited distinctiveness for robust material classification.

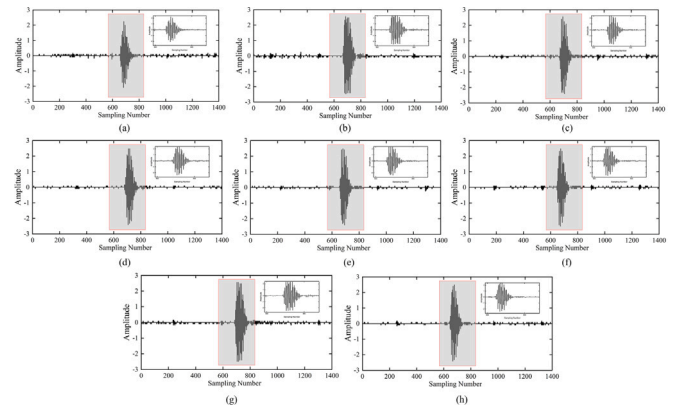


Fig. 9. Time-domain waveform of ultrasonic echo signals from different materials at a distance of 45 cm dataset constructed in this study. (a) is foam, (b) is wood, (c) is paper, (d) is copper, (e) is stone, (f) is iron, (g) is aluminum, (h) is cloth.

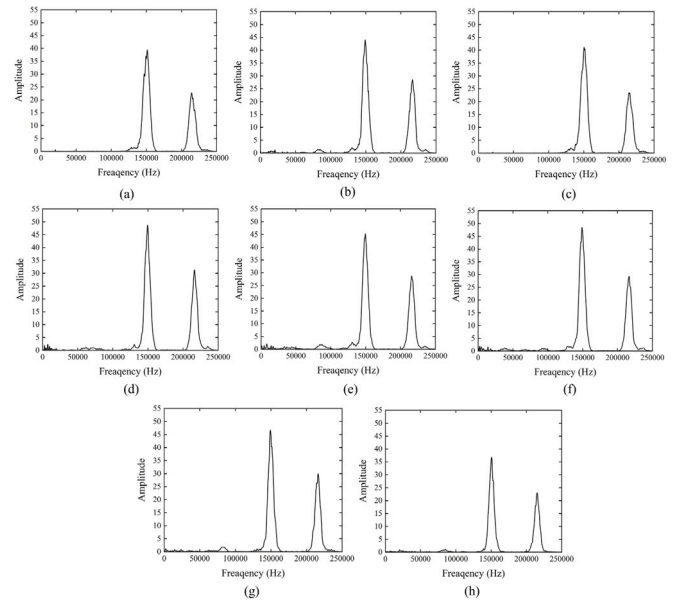


Fig. 10. Frequency-domain spectrum of ultrasonic echo signals from different materials at a distance of 45 cm dataset constructed in this study. (a) is foam, (b) is wood, (c) is paper, (d) is copper, (e) is stone, (f) is iron, (g) is aluminum, (h) is cloth.

Taken together, the experimental data indicate that conventional signal features in the time and frequency domains are insufficient for accurate material identification. Therefore, advanced techniques are required to recognize and interpret these subtle differences in order to extract comprehensive characteristics about material properties, spatial structure, temporal dynamics, and more. These approaches are essential for inferring detailed material attributes and for enabling precise classification in real-world robotic perception tasks.

4.2.2. Data augmentation

Given the limited size of the experimental dataset, there is a significant risk of model overfitting. To address this issue, data augmentation techniques are employed to increase data diversity and improve the generalization ability of the model. These techniques typically generate synthetic training data through controlled random transformations. Existing studies have proposed a variety of time-series data augmentation methods, including window slicing [54], jittering or flipping [55,56],

scaling [56,57], time warping [58], and frequency warping [57]. These methods have proven effective in enhancing model robustness across a wide range of signal processing tasks.

In this study, a customized data augmentation approach tailored for ultrasonic echo signals is introduced, referred to as EMD–WGN. This method combines Empirical Mode Decomposition (EMD) with additive White Gaussian Noise (WGN) to generate realistic and diverse synthetic data. The procedure consists of the following steps. First, the ultrasonic echo signal is decomposed using EMD to obtain a set of Intrinsic Mode Function (IMF) components and a residual term. Among these, the first IMF component (IMF1) is selected as the target for augmentation. Subsequently, multiple sequences of white Gaussian noise are generated with varying levels of intensity to simulate real-world acoustic interference and sensor noise. Finally, the generated noise is added to the IMF1 component with different scaling factors to synthesize new signal variants. The resulting augmented signals maintain the essential characteristics of the original data while introducing controlled variability, thereby improving the robustness and generalizability of the model. This EMD–WGN augmentation strategy leverages the adaptive decomposition capability of EMD and the stochastic nature of WGN to effectively expand the dataset in a structure-preserving manner. It is particularly suitable for applications involving limited samples and subtle signal variations.

4.2.3. System implementation for robotic deployment

To validate the end-to-end applicability of the proposed AE-CS-TCN, the trained model was deployed on a mobile robotic platform equipped with the ultrasonic sensing system described before. The entire perception pipeline integrates signal emission, echo acquisition, real-time inference, and classification feedback within a self-contained system operating entirely onboard. The hardware setup consists of an STM32-based ultrasonic control board, a pair of 200 kHz transducers for transmission and reception, and an industrial PC acting as the computation unit. The STM32 module generates precisely timed excitation pulses and synchronizes the reception window through hardware interrupts, while the received echoes are amplified and digitized by a 16-bit ADC at a 2 MHz sampling rate before being streamed to the onboard processor. The AE-CS-TCN model, implemented in PyTorch and optimized, performs direct inference on raw waveforms through a ROS-based framework that connects the sensing and motion-control modules. Each processing cycle includes pulse triggering, echo synchronization, signal normalization, forward propagation through the network, and real-time transmission of the predicted material label to the robot's control node. The total latency from signal emission to classification output is approximately 35–40 ms, satisfying the real-time perception requirements for mobile navigation. Several engineering challenges were encountered during deployment. Temporal misalignment between the transmitting and receiving modules initially caused unstable inference, which was resolved by introducing a dedicated synchronization line to ensure microsecond-level timing accuracy. Mechanical vibration and acoustic interference from the robot's motion system degraded signal quality; this issue was mitigated through vibration isolation of the transducers and adaptive noise filtering. Furthermore, real-time inference required efficient computational optimization. After profiling and model compression, AE-CS-TCN achieved 25 frames per second on an Intel i7 CPU with less than 15% utilization. These results confirm that the proposed model operates robustly as an autonomous ultrasonic perception module, capable of real-time, non-contact material recognition in visually degraded environments, thus bridging the gap between deep learning algorithm design and practical robotic deployment.

5. Results and discussion

5.1. Metrics

The confusion matrix is a fundamental tool used to tabulate and analyze the relationship between the true class labels and the predicted

class labels produced by a classification model. It provides a detailed visualization of the distribution of correctly and incorrectly classified instances across different categories. The confusion matrix is composed of four components: True Positive (T_P), which counts the number of positive instances correctly predicted as positive; False Negative (F_N), representing positive instances incorrectly predicted as negative, indicating missed detections; False Positive (F_P), corresponding to negative instances incorrectly predicted as positive, which are false alarms; and True Negative (T_N), which is the number of negative instances correctly classified as negative.

These quantities form the basis for calculating various evaluation metrics that quantitatively assess the performance of the classification model. One of the most straightforward metrics is accuracy, which represents the ratio of correctly classified samples (including both true positives and true negatives) to the total number of samples. Accuracy reflects the overall ability of the model to correctly recognize and classify unknown materials, and is expressed mathematically as:

$$Accuracy = (T_P + T_N) / (T_P + F_N + F_P + T_N) \quad (9)$$

Although accuracy provides a general sense of model performance, it may be misleading in cases where the dataset is imbalanced, as it does not differentiate between types of classification errors. Therefore, more nuanced metrics such as precision and recall are commonly used to better characterize model behavior. Precision measures the proportion of true positive predictions out of all samples predicted as positive, reflecting the model's ability to minimize false positives and thus the reliability of positive predictions. Precision is given by:

$$P = T_P / (T_P + F_P) \quad (10)$$

Recall, also known as sensitivity or true positive rate, quantifies the proportion of actual positive cases that are correctly identified by the model. It indicates how effectively the model captures all relevant positive instances, thus minimizing false negatives. Recall is formulated as:

$$R = T_P / (T_P + F_N) \quad (11)$$

In practical applications, there often exists a trade-off between precision and recall; a model optimized for one may degrade the other. To provide a balanced assessment, the $F1$ score is employed as the harmonic mean of precision and recall, combining these two metrics into a single value. A higher $F1$ score indicates a better overall performance in correctly recognizing the target materials while controlling both false positives and false negatives. The $F1$ score is calculated as:

$$F1 = (2 \times P \times R) / (P + R) \quad (12)$$

5.2. Model training parameter settings

The detailed network structure of the proposed AE-CS-TCN model is summarized in Table 2, which outlines the configuration of each network layer, including input and output dimensions across different processing branches. For model optimization, the cross-entropy loss function is minimized using the Adam optimizer with an initial learning rate of 0.0001, a batch size of 64, and a dropout rate of 0.5. The selection of these hyperparameters is guided by a grid search strategy to achieve a balance between convergence stability and generalization performance. The cross-scale block employs scale control parameters of 4 and 6 to enhance multi-scale feature representation, while the temporal convolution blocks adopt dilation factors of [1, 2, 4, 8] and [8, 16, 32, 64] to capture short- and long-term temporal dependencies, respectively. Batch normalization is applied during preprocessing to standardize input features and accelerate training convergence. The dataset is divided into 80% for training and 20% for testing using stratified sampling to maintain class distribution consistency.

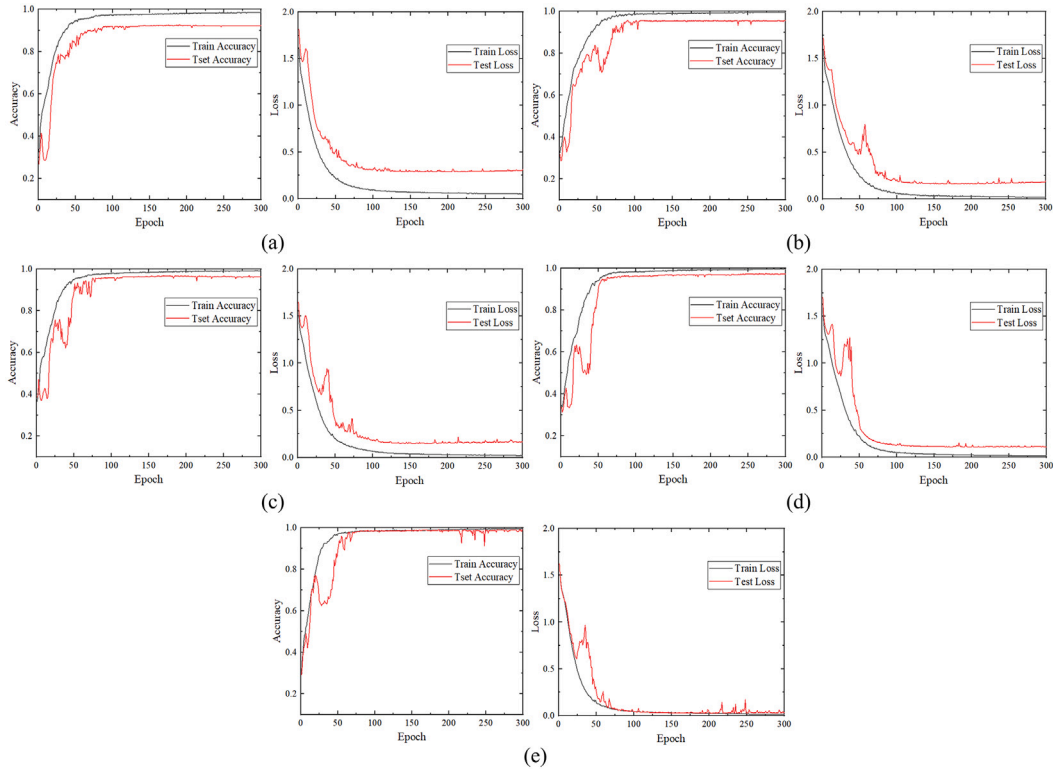


Fig. 11. Accuracy and Loss plot for dataset expanded after different data augmentation methods. (a) is the accuracy and loss plot for the original dataset, (b) is the accuracy and loss plot for the dataset expanded after Scaling, (c) is the dataset expanded after Jittering, (d) is the accuracy and loss plot for the dataset expanded after Flipping, (e) is the accuracy and loss plot for the dataset expanded after EMD-WGN.

Table 3

Evaluation results of different data augmentation methods at a distance of 45 cm dataset constructed in this study.

Method	Metrics			
	Accuracy (Train)	Loss (Train)	Accuracy (Test)	Loss (Test)
Original dataset	0.986 ± 0.006	0.0561 ± 0.009	0.922 ± 0.009	0.312 ± 0.007
Scaling	0.995 ± 0.004	0.0181 ± 0.010	0.956 ± 0.005	0.196 ± 0.008
Jittering	0.992 ± 0.007	0.0248 ± 0.011	0.962 ± 0.004	0.165 ± 0.011
Flipping	0.994 ± 0.005	0.0208 ± 0.010	0.970 ± 0.006	0.132 ± 0.010
EMD-WGN	0.996 ± 0.004	0.0157 ± 0.012	0.977 ± 0.005	0.0745 ± 0.010

5.3. Comparison results of EMD-WGN with other data augmentation methods

To validate the effectiveness of the EMD-WGN data augmentation method, it is compared with several common data augmentation techniques, including jittering, flipping, and scaling. Comparative experiments are conducted using an echo dataset collected at a 45 cm distance, with the dataset size expanded from 1600 to 12800 samples. Each experiment is repeated five times to ensure statistical robustness, and the results are reported as mean values with corresponding standard deviations to quantify result variability. The AE-CS-TCN model serves as the test model, with performance evaluated using two key metrics: accuracy (measuring classification correctness) and loss value (quantifying prediction error). The training and testing results are visualized in Fig. 11 and summarized in Table 3.

Table 3 presents the evaluation results of the various data augmentation methods. Among all methods, EMD-WGN yields the highest test accuracy at 0.977 ± 0.005 , outperforming scaling (0.956 ± 0.005), jittering (0.962 ± 0.004), and flipping (0.970 ± 0.006). It also achieves the lowest test loss (0.0745 ± 0.010), indicating more stable and reliable predictions. A brief analysis of the table reveals that all data augmentation methods improve model performance compared to the

original dataset. For the original dataset, the gap between training accuracy (0.986 ± 0.006) and test accuracy (0.922 ± 0.009) is notable, accompanied by a high test loss (0.312 ± 0.007), suggesting overfitting. In contrast, data augmentation narrows this gap, with EMD-WGN showing the smallest discrepancy (training accuracy: 0.996 ± 0.004 ; test accuracy: 0.977 ± 0.005), indicating enhanced generalization. The results indicate that EMD-WGN enhanced data most effectively aligns with the distribution of real-world signals, enabling the model to learn generalized features rather than memorizing training specifics. The low standard deviations across all metrics confirm the consistency and reproducibility of the experimental results. Unlike basic random transformations, EMD-WGN simulates the intrinsic signal characteristics of actual materials, providing a richer and more realistic training environment. This explains its superior performance in improving both accuracy and generalization ability, effectively mitigating overfitting.

5.4. The results of the proposed model under different distances

To comprehensively evaluate the performance of the proposed model across varying distances, material recognition experiments are conducted on datasets collected at 30 cm, 35 cm, 40 cm, 45 cm, and 50 cm, as well as on a combined dataset integrating samples from all

Table 4
Evaluation results of the AE-CS-TCN model under different distance datasets constructed in this study.

Dataset	Metrics				
	Accuracy	Loss	Precision	Recall	F1-score
30 cm	0.960 ± 0.003	0.1202 ± 0.007	0.954 ± 0.008	0.962 ± 0.005	0.960 ± 0.008
35 cm	0.964 ± 0.004	0.1102 ± 0.008	0.961 ± 0.007	0.964 ± 0.006	0.963 ± 0.008
40 cm	0.970 ± 0.005	0.0921 ± 0.009	0.969 ± 0.006	0.965 ± 0.008	0.967 ± 0.007
45 cm	0.977 ± 0.005	0.0745 ± 0.010	0.978 ± 0.005	0.972 ± 0.007	0.974 ± 0.008
50 cm	0.971 ± 0.004	0.0902 ± 0.011	0.970 ± 0.008	0.965 ± 0.006	0.968 ± 0.006
Mixed	0.967 ± 0.003	0.1023 ± 0.010	0.962 ± 0.005	0.971 ± 0.007	0.966 ± 0.006

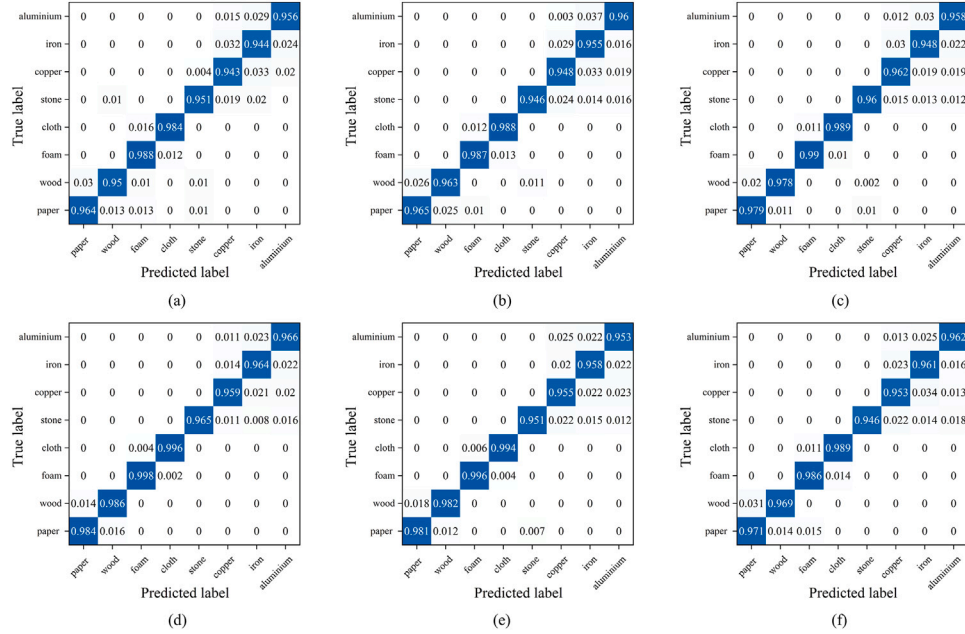


Fig. 12. Confusion matrix for the AE-CS-TCN model under different distance datasets constructed in this study. (a) is confusion matrix at 30 cm dataset, (b) is confusion matrix at 35 cm dataset, (c) is confusion matrix at 40 cm dataset, (d) is confusion matrix at 45 cm dataset, (e) is confusion matrix at 50 cm dataset, (f) is confusion matrix at mixed dataset.

these distances. The EMD-WGN data augmentation method is applied to the datasets at these varying distances, followed by comparative experiments using the AE-CS-TCN model with consistent parameters. Each experiment is repeated five times to ensure statistical robustness, and results are reported as mean values with standard deviations to quantify performance stability. The results of these experiments are detailed in Table 4 and Fig. 12.

The 30 cm dataset shows the lowest performance among single-distance datasets, with the lowest accuracy (0.960 ± 0.003) and highest loss (0.1202 ± 0.007). This is attributed to the near-field effects at close range, where overlapping echoes from the material surface and edges create complex interference patterns, obscuring fine-grained features. In contrast, the 50 cm dataset is affected by signal attenuation: as distance increases, ultrasonic energy dissipation leads to reduced echo amplitude and blurred waveform details, resulting in slightly lower precision (0.970 ± 0.008) compared to 45 cm (0.978 ± 0.005). Notably, the standard deviation observed across all evaluated metrics did not exceed 0.011, demonstrating consistent model performance over five experimental repetitions.

An evaluation conducted on the mixed dataset, which aggregates samples obtained at varying distances, demonstrates a high classification accuracy of 0.967 ± 0.003 and a well-balanced F1-score of 0.966 ± 0.006 . In contrast to fixed-distance benchmarks, this mixed dataset reflects the continuous variability in robot-to-target distance that occurs in real-world environments. These variations arise from factors such as vehicle movement over uneven terrain, wheel slip, and dynamic changes in target location. Importantly, the model's performance remains within 1.0% of its peak accuracy at the 45 cm

optimal distance, indicating a high degree of robustness to environmental fluctuations without the need for frequent recalibration. This level of resilience is particularly critical for autonomous systems, where consistent material classification and reliable decision-making must be maintained despite dynamically changing conditions.

The model shows higher recognition accuracy for foam, paper, wood, and cloth, but lower accuracy for iron, stone, copper, and aluminum. This is likely because the acoustic impedance differences among iron, stone, copper, and aluminum are minimal (as inferred from the theoretical acoustic model and Refs. [7]), making them less distinguishable compared to other materials. However, even for these challenging materials, the model maintains an F1-score above 0.95 across all distances, confirming its robustness. The propagation characteristics of ultrasonic signals vary with distance, influencing the strength and clarity of ultrasonic echo signals, but the model's consistent performance across the mixed dataset demonstrates its ability to adapt to these variations, laying a foundation for practical robotic material recognition systems.

5.5. Ablation study

This section focuses on evaluating the effectiveness of the spatial attention block, temporal convolution block, cross scale block, and cross attention block, and their influence on the performance of the AE-CS-TCN model. Five variations of the model are constructed: one without the spatial attention block (V1), one without the temporal convolution block (V2), one without the cross scale block (V3), one without the

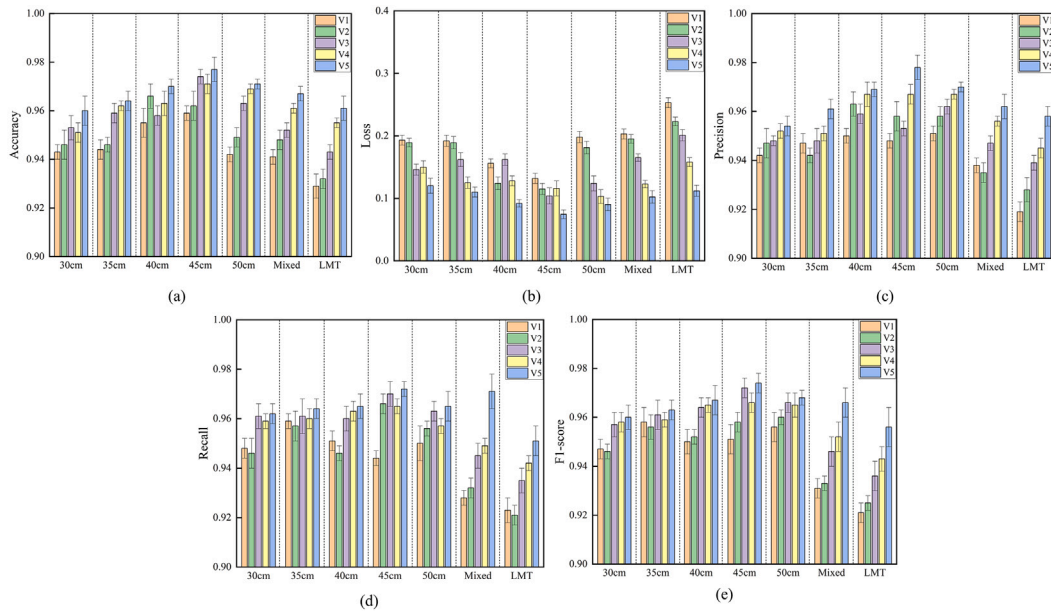


Fig. 13. Experiment results of ablation study under different datasets. (a) is accuracy for different versions of the model, (b) is loss for different versions of the model, (c) is precision for different versions of the model, (d) is recall for different versions of the model, (e) is F1-score for different versions of the model.

cross attention block (V4), and the complete version (V5). Comparative experiments are conducted using these five model variations, with each experiment repeated five times to ensure statistical robustness. Results are reported as mean values with standard deviations, as visualized in Fig. 13, to validate the effectiveness of the key components in the AE-CS-TCN model.

Fig. 13 presents the comparative performance evaluation of the five model variants across datasets captured at varying distances. The experimental results demonstrate that the complete AE-CS-TCN (V5) achieves the highest average recognition accuracy, outperforming all ablated versions in material classification tasks. This confirms the collective importance of the proposed architectural components (spatial attention block, temporal convolution block, cross-scale block, and cross attention block) as the removal of any single block leads to a measurable degradation in recognition performance. Notably, the spatial attention block contributes most significantly to model accuracy, emphasizing its role in enhancing discriminative feature extraction. The temporal convolution block also exhibits a strong influence, validating its effectiveness in capturing sequential dependencies. While the cross scale block provides a moderate performance boost through multi-resolution feature fusion, the cross attention block, though beneficial, has a comparatively smaller impact, suggesting its role as a supplementary refinement module rather than a primary driver of recognition accuracy.

To evaluate the performance of the different model versions, we employed the t-SNE algorithm [59] to visualize the material features extracted from the final fully connected (FC) layer of each model. The visualization results on different datasets are presented in Figs. 14 and 15. As observed in these figures, the V1 and V2 models show a certain degree of clustering tendency among material features; however, significant overlaps between different material classes limit the extent of separation. In contrast, the V3 and V4 models exhibit a more pronounced clustering behavior, indicating improved feature discriminability across different material categories. Notably, the V5 model achieves the best clustering performance, with clearly defined and well-separated material clusters. This enhanced feature separation contributes to a higher recognition accuracy, demonstrating the effectiveness of the model improvements introduced in V5.

The spatial attention block leverages two complementary feature descriptors (average pooling and max pooling) to capture spatial structural information at multiple scales, derived from both the time domain

and the frequency domain. The resulting feature maps are then integrated to generate a spatial attention map X_s , which dynamically assigns different attention weights to distinct spatial regions (e.g., the 125 kHz–250 kHz and 0–125 kHz frequency bands illustrated in Fig. 10). This block enables the model to emphasize informative spatial structures while suppressing irrelevant or redundant components, thereby significantly improving the accuracy of material recognition. Complementing the spatial attention block, the temporal convolution block utilizes dilated causal convolutions to effectively model long-term temporal dependencies in ultrasound echo signals sampled at 1400 points. By progressively increasing the dilation factors ([1, 2, 4, 8] in the shallow layers and [8, 16, 32, 64] in the deeper layers), the receptive field size expands exponentially to 2^d times the initial size, where d denotes the layer depth. This design ensures that the model can capture contextual information over extended temporal ranges without incurring excessive computational overhead or violating the causality constraint, which is critical for processing sequential sensor data in robotic applications. Furthermore, the cross scale block introduces a hierarchical residual-like fusion strategy that integrates multi-scale representations using scale control parameters set to 4 and 6. This block enhances the model's ability to process complex time-series data by preserving fine-grained details while simultaneously capturing high-level abstractions. The cross scale fusion mechanism significantly improves the model's capacity to recognize intricate material features across a wide range of frequencies. Although the performance improvement attributed to the cross attention block may appear modest in isolation, its functional role is pivotal in integrating multidimensional feature representations. By synergistically combining different attention mechanisms, this block facilitates the fusion of semantic and contextual information, enabling the model to focus on critical segments of the echo signals. It also reveals latent semantic relationships among various signal components, thereby deepening the model's understanding of inter-component dependencies and further strengthening its material recognition capability. Extensive experimental results confirm that each of these architectural components plays a vital role in achieving optimal performance within the AE-CS-TCN framework. Ablation studies not only validate the individual effectiveness of each block but also underscore the importance of their synergistic integration in delivering robust and accurate material classification across diverse sensing scenarios.

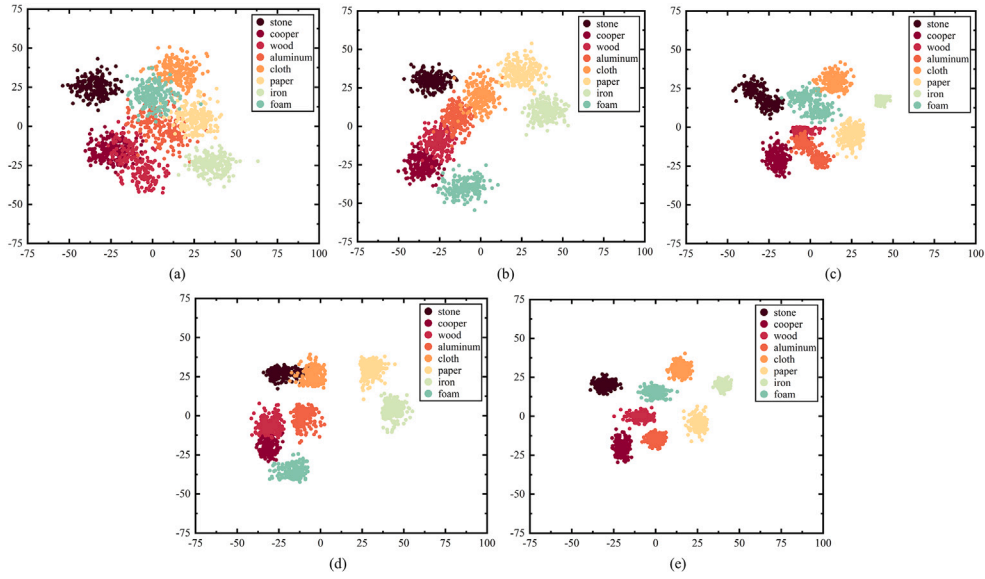


Fig. 14. t-SNE results for the AE-CS-TCN model in different versions at a distance of 45 cm dataset constructed in this study. (a) is t-SNE for V1, (b) is t-SNE for V2, (c) is t-SNE for V3, (d) is t-SNE for V4, (e) is t-SNE for V5.

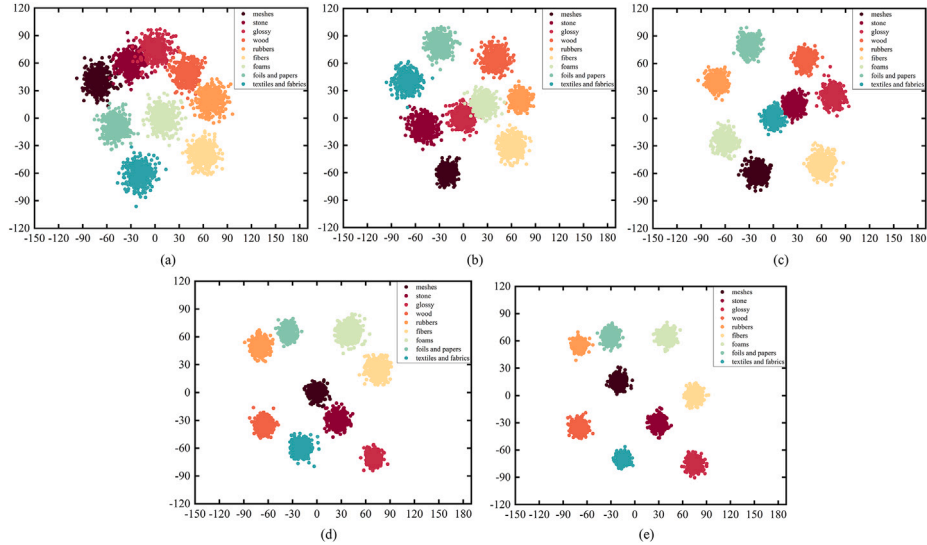


Fig. 15. t-SNE results for the AE-CS-TCN model in different versions at LMT dataset. (a) is t-SNE for V1, (b) is t-SNE for V2, (c) is t-SNE for V3, (d) is t-SNE for V4, (e) is t-SNE for V5.

Table 5 presents the material recognition accuracies of different versions of the AE-CS-TCN model across various distances dataset constructed in this study. At shorter distances (30 cm–40 cm), V1 and V2 show lower accuracy on metallic materials such as aluminum, copper, and iron, suggesting that spatial attention and temporal convolution are important for capturing fine surface details and temporal patterns. As distance increases, performance of all models slightly drops, but V5 consistently outperforms the others, especially on complex materials like cloth, foam, and wood. Notably, V5 achieves the highest accuracy on foam and shows improved recognition on cloth and wood, indicating that the full model effectively integrates spatial, temporal, and cross-scale features. In contrast, models missing key blocks (for example, V1 without spatial attention) perform worse on materials requiring detailed surface analysis. Table 6 shows the material recognition accuracies of different model variants (V1 to V5) on the LMT dataset, which includes nine material categories such as Meshes, Stone, Glossy, Wood, Textiles and Fabrics, and so on. The results show that V5 achieves the highest accuracy across most material categories. For example,

V5 reaches nearly perfect recognition on Foams (0.998) and shows strong performance on Wood (0.985) and Fibers (0.986), indicating that the full model effectively captures both spatial and temporal features necessary for recognizing complex materials. In contrast, V1 (without spatial attention) shows the lowest accuracy on Fibers (0.921) and Foils and Papers (0.903), demonstrating that spatial attention is crucial for recognizing fine-grained textures. V2 (without temporal convolution) performs worse on Wood (0.941) and Meshes (0.935), suggesting that temporal modeling helps capture sequential patterns in the data. Models V4 and V5 also show strong performance on Foams and Wood, highlighting the importance of cross attention and full model integration. The Glossy category shows similar accuracy across all models, indicating that it relies on more general visual features.

5.6. Comparison results of the AE-CS-TCN model with other models

To assess the performance of the AE-CS-TCN model in material recognition, its experimental results are compared with those of other

Table 5

The accuracy of different materials of ablation study at different distance datasets constructed in this study.

Dataset	Models	Accuracy							
		Paper	Cloth	Foam	Wood	Stone	Aluminum	Copper	Iron
30 cm	V1	0.953	0.986	0.992	0.948	0.912	0.920	0.921	0.913
	V2	0.949	0.988	0.989	0.941	0.922	0.925	0.927	0.926
	V3	0.962	0.984	0.993	0.944	0.932	0.935	0.936	0.938
	V4	0.961	0.992	0.989	0.947	0.93	0.931	0.933	0.925
	V5	0.964	0.984	0.988	0.950	0.951	0.956	0.943	0.944
35 cm	V1	0.965	0.992	0.982	0.943	0.924	0.910	0.916	0.921
	V2	0.958	0.984	0.990	0.928	0.932	0.924	0.930	0.922
	V3	0.970	0.995	0.993	0.953	0.939	0.942	0.936	0.940
	V4	0.972	0.990	0.991	0.966	0.940	0.939	0.942	0.948
	V5	0.965	0.988	0.987	0.963	0.946	0.960	0.948	0.955
40 cm	V1	0.952	0.992	0.989	0.941	0.935	0.938	0.948	0.945
	V2	0.989	0.994	0.996	0.969	0.948	0.942	0.947	0.943
	V3	0.971	0.995	0.988	0.959	0.941	0.940	0.937	0.934
	V4	0.968	0.990	0.987	0.965	0.946	0.948	0.948	0.950
	V5	0.979	0.989	0.990	0.978	0.960	0.958	0.962	0.948
45 cm	V1	0.969	0.996	0.998	0.958	0.937	0.942	0.939	0.936
	V2	0.972	0.993	0.996	0.957	0.955	0.948	0.935	0.941
	V3	0.992	0.995	0.997	0.966	0.968	0.953	0.959	0.962
	V4	0.988	0.992	0.994	0.977	0.943	0.959	0.955	0.957
	V5	0.984	0.996	0.998	0.986	0.965	0.966	0.959	0.964
50 cm	V1	0.948	0.991	0.988	0.934	0.923	0.916	0.921	0.914
	V2	0.941	0.998	0.994	0.931	0.934	0.935	0.929	0.932
	V3	0.966	0.996	0.992	0.952	0.944	0.950	0.948	0.954
	V4	0.982	0.993	0.995	0.967	0.952	0.949	0.956	0.955
	V5	0.981	0.994	0.996	0.982	0.951	0.953	0.955	0.958
Mixed	V1	0.952	0.991	0.982	0.914	0.921	0.922	0.925	0.923
	V2	0.941	0.995	0.994	0.931	0.934	0.930	0.929	0.932
	V3	0.947	0.989	0.990	0.945	0.934	0.941	0.939	0.935
	V4	0.954	0.991	0.992	0.953	0.954	0.945	0.951	0.950
	V5	0.971	0.989	0.986	0.969	0.946	0.962	0.953	0.961

Table 6

The accuracy of different materials of ablation study at LMT dataset.

Dataset	Models	Accuracy								
		Meshes	Stone	Glossy	Wood	Rubbers	Fibers	Foams	Foils and Papers	Textiles and Fabrics
LMT	V1	0.943	0.921	0.941	0.938	0.912	0.921	0.972	0.903	0.912
	V2	0.935	0.951	0.932	0.941	0.911	0.932	0.968	0.909	0.911
	V3	0.942	0.974	0.953	0.954	0.932	0.937	0.965	0.912	0.915
	V4	0.975	0.981	0.952	0.978	0.935	0.941	0.989	0.925	0.919
	V5	0.975	0.982	0.952	0.985	0.946	0.986	0.998	0.915	0.913

state-of-the-art models, including KNN [31], Decision Tree [39], SVM [7], CNN [36], LSTM [60], GRU [61], CNN-LSTM [62], and TCN [63]. For a fair and rigorous comparison, all models are trained and evaluated using the same datasets, with the parameters of each model adopted from their respective original implementations. Each experiment is repeated five times to account for random initialization effects, and results are reported as mean values with standard deviations to ensure statistical reliability. The comparative experimental results are presented in Table 7, Table 8, and Table 9.

The experimental results presented in Table 7 demonstrate that the proposed AE-CS-TCN model consistently outperforms all baseline methods across multiple evaluation scenarios, including varying distances (30 cm to 50 cm), mixed-distance conditions, and the LMT dataset. The model achieves the highest accuracy and F1-score, highlighting its robustness in handling diverse material recognition tasks. Notably, the AE-CS-TCN exhibits significant improvements over traditional machine learning methods (e.g., KNN, SVM) and deep learning baselines (e.g., CNN, LSTM, GRU), particularly in complex scenarios such as mixed-distance testing (accuracy: 0.967 ± 0.003). Furthermore, the AE-CS-TCN model has a testing time of only 0.5 s, a speed that fully meets real-time requirements. This endows it with significant practical value in real-world application scenarios such as mobile robots that demand rapid responses, further highlighting the model's advantage

in real-time performance while maintaining high-precision recognition. This superior performance can be attributed to the effective integration by the model of multi-scale spatiotemporal feature extraction and attention mechanisms, which enhance discriminative capability for fine-grained materials (e.g., foams and textiles). In contrast, conventional approaches suffer from limitations in either spatial modeling (CNN), long-term dependency capture (LSTM/GRU), or cross-scale feature fusion (TCN). The consistent accuracy of the AE-CS-TCN (standard deviation ≤ 0.008) further confirms the stability of the model across varying conditions.

The experimental results presented in Tables 8 and 9 demonstrate that the proposed AE-CS-TCN model achieves state-of-the-art performance in material recognition across multiple testing scenarios. As shown in Table 8, the model consistently outperforms all baseline methods at varying distances (30 cm to 50 cm), with particularly impressive results for foam recognition (0.998 accuracy at 45 cm) and metallic materials (0.956 for aluminum at 35 cm). The performance advantage becomes more pronounced at longer distances, where the proposed model maintains above 0.95 accuracy while traditional methods like KNN drop below 0.85, demonstrating superior robustness to distance variations. Table 9 further validates these findings on the LMT dataset, where the model achieves near perfect accuracy for foams (0.998) and maintains strong performance across all material categories, including challenging textiles (0.913) and foils (0.915). The results suggest

Table 7
Evaluation results of the AE-CS-TCN model and other models.

Dataset	Models	Metrics			
		Accuracy	Precision	Recall	F1-score
30 cm	KNN	0.861 ± 0.005	0.857 ± 0.008	0.859 ± 0.005	0.858 ± 0.006
	Decision Tree	0.867 ± 0.007	0.862 ± 0.007	0.858 ± 0.009	0.861 ± 0.010
	SVM	0.871 ± 0.005	0.867 ± 0.005	0.861 ± 0.007	0.864 ± 0.009
	CNN	0.876 ± 0.010	0.865 ± 0.007	0.857 ± 0.009	0.862 ± 0.011
	LSTM	0.893 ± 0.009	0.891 ± 0.011	0.882 ± 0.009	0.886 ± 0.010
	GRU	0.912 ± 0.008	0.908 ± 0.010	0.902 ± 0.008	0.905 ± 0.006
	CNN-LSTM	0.926 ± 0.006	0.918 ± 0.008	0.907 ± 0.007	0.914 ± 0.009
	TCN	0.945 ± 0.008	0.935 ± 0.006	0.928 ± 0.005	0.931 ± 0.004
	Proposed	0.960 ± 0.003	0.954 ± 0.008	0.962 ± 0.005	0.960 ± 0.008
35 cm	KNN	0.863 ± 0.005	0.856 ± 0.009	0.860 ± 0.006	0.858 ± 0.006
	Decision Tree	0.866 ± 0.007	0.863 ± 0.010	0.856 ± 0.009	0.861 ± 0.010
	SVM	0.873 ± 0.004	0.865 ± 0.005	0.861 ± 0.007	0.863 ± 0.008
	CNN	0.882 ± 0.006	0.876 ± 0.008	0.881 ± 0.006	0.878 ± 0.010
	LSTM	0.891 ± 0.005	0.894 ± 0.009	0.885 ± 0.008	0.890 ± 0.006
	GRU	0.904 ± 0.006	0.889 ± 0.007	0.898 ± 0.006	0.894 ± 0.005
	CNN-LSTM	0.915 ± 0.005	0.905 ± 0.008	0.910 ± 0.009	0.908 ± 0.006
	TCN	0.948 ± 0.006	0.937 ± 0.009	0.923 ± 0.007	0.931 ± 0.008
	Proposed	0.964 ± 0.004	0.961 ± 0.007	0.964 ± 0.006	0.963 ± 0.008
40 cm	KNN	0.863 ± 0.006	0.857 ± 0.008	0.863 ± 0.006	0.861 ± 0.006
	Decision Tree	0.868 ± 0.004	0.863 ± 0.010	0.856 ± 0.009	0.861 ± 0.010
	SVM	0.875 ± 0.006	0.865 ± 0.005	0.861 ± 0.006	0.863 ± 0.008
	CNN	0.883 ± 0.006	0.892 ± 0.006	0.874 ± 0.007	0.882 ± 0.005
	LSTM	0.893 ± 0.005	0.885 ± 0.008	0.891 ± 0.006	0.889 ± 0.006
	GRU	0.898 ± 0.006	0.894 ± 0.005	0.885 ± 0.007	0.891 ± 0.006
	CNN-LSTM	0.920 ± 0.007	0.908 ± 0.008	0.921 ± 0.008	0.913 ± 0.009
	TCN	0.950 ± 0.006	0.935 ± 0.007	0.948 ± 0.006	0.940 ± 0.008
	Proposed	0.970 ± 0.005	0.969 ± 0.006	0.965 ± 0.008	0.967 ± 0.007
45 cm	KNN	0.858 ± 0.005	0.857 ± 0.008	0.853 ± 0.005	0.856 ± 0.006
	Decision Tree	0.865 ± 0.007	0.861 ± 0.007	0.855 ± 0.009	0.859 ± 0.010
	SVM	0.870 ± 0.005	0.868 ± 0.005	0.861 ± 0.007	0.865 ± 0.009
	CNN	0.877 ± 0.007	0.874 ± 0.008	0.868 ± 0.009	0.872 ± 0.006
	LSTM	0.889 ± 0.005	0.890 ± 0.009	0.879 ± 0.008	0.884 ± 0.007
	GRU	0.908 ± 0.006	0.912 ± 0.008	0.901 ± 0.007	0.907 ± 0.006
	CNN-LSTM	0.921 ± 0.005	0.903 ± 0.007	0.920 ± 0.006	0.914 ± 0.005
	TCN	0.952 ± 0.007	0.924 ± 0.007	0.942 ± 0.008	0.932 ± 0.005
	Proposed	0.977 ± 0.005	0.978 ± 0.005	0.972 ± 0.007	0.974 ± 0.008
50 cm	KNN	0.861 ± 0.005	0.854 ± 0.009	0.860 ± 0.007	0.857 ± 0.006
	Decision Tree	0.868 ± 0.007	0.865 ± 0.008	0.856 ± 0.009	0.862 ± 0.009
	SVM	0.875 ± 0.004	0.867 ± 0.005	0.861 ± 0.007	0.864 ± 0.008
	CNN	0.882 ± 0.008	0.892 ± 0.007	0.875 ± 0.008	0.884 ± 0.006
	LSTM	0.892 ± 0.005	0.887 ± 0.008	0.878 ± 0.007	0.884 ± 0.006
	GRU	0.905 ± 0.006	0.891 ± 0.007	0.904 ± 0.005	0.897 ± 0.006
	CNN-LSTM	0.916 ± 0.008	0.908 ± 0.009	0.918 ± 0.008	0.912 ± 0.006
	TCN	0.951 ± 0.005	0.948 ± 0.007	0.935 ± 0.006	0.941 ± 0.005
	Proposed	0.971 ± 0.004	0.970 ± 0.008	0.965 ± 0.006	0.968 ± 0.006
Mixed	KNN	0.855 ± 0.006	0.854 ± 0.009	0.848 ± 0.008	0.851 ± 0.006
	Decision Tree	0.862 ± 0.004	0.860 ± 0.007	0.856 ± 0.009	0.857 ± 0.008
	SVM	0.871 ± 0.005	0.865 ± 0.005	0.861 ± 0.006	0.863 ± 0.008
	CNN	0.879 ± 0.006	0.872 ± 0.009	0.880 ± 0.008	0.875 ± 0.007
	LSTM	0.895 ± 0.005	0.883 ± 0.007	0.892 ± 0.006	0.889 ± 0.005
	GRU	0.908 ± 0.006	0.892 ± 0.008	0.899 ± 0.009	0.895 ± 0.008
	CNN-LSTM	0.923 ± 0.007	0.915 ± 0.008	0.921 ± 0.007	0.918 ± 0.006
	TCN	0.949 ± 0.005	0.939 ± 0.007	0.921 ± 0.006	0.933 ± 0.005
	Proposed	0.967 ± 0.003	0.962 ± 0.005	0.971 ± 0.007	0.966 ± 0.006
LMT	KNN	0.868 ± 0.005	0.862 ± 0.008	0.858 ± 0.005	0.861 ± 0.006
	Decision Tree	0.875 ± 0.007	0.871 ± 0.007	0.865 ± 0.009	0.868 ± 0.010
	SVM	0.881 ± 0.005	0.878 ± 0.005	0.871 ± 0.007	0.875 ± 0.009
	CNN	0.892 ± 0.007	0.879 ± 0.008	0.882 ± 0.007	0.880 ± 0.006
	LSTM	0.906 ± 0.006	0.881 ± 0.007	0.895 ± 0.006	0.890 ± 0.005
	GRU	0.915 ± 0.007	0.905 ± 0.009	0.899 ± 0.007	0.901 ± 0.006
	CNN-LSTM	0.942 ± 0.006	0.926 ± 0.006	0.918 ± 0.007	0.922 ± 0.005
	TCN	0.952 ± 0.005	0.945 ± 0.007	0.938 ± 0.006	0.939 ± 0.008
	Proposed	0.961 ± 0.005	0.958 ± 0.004	0.951 ± 0.006	0.956 ± 0.008

that the model's architectural innovations, particularly in multi-scale feature integration and attention mechanisms, effectively address key challenges in material recognition, such as fine texture discrimination (evidenced by 0.986 accuracy for fibers) and reflective surface handling (0.956 for metals). Compared to conventional approaches, the proposed solution shows significant improvements in both absolute performance

and consistency (minimal performance degradation across distance variations), making it particularly suitable for real-world applications where environmental conditions may vary.

The superior performance of the proposed model in comparison to conventional classifiers such as KNN, SVM, and Decision Tree stems primarily from its capacity to extract high-level abstractions directly

Table 8

The accuracy of different materials for the AE-CS-TCN model and other models at different distance datasets constructed in this study.

Dataset	Models	Accuracy							
		Paper	Cloth	Foam	Wood	Stone	Aluminum	Copper	Iron
30 cm	KNN	0.858	0.913	0.918	0.858	0.842	0.834	0.835	0.832
	Decision Tree	0.859	0.925	0.93	0.859	0.846	0.838	0.839	0.839
	SVM	0.861	0.928	0.932	0.869	0.848	0.839	0.842	0.845
	CNN	0.872	0.929	0.932	0.886	0.852	0.847	0.843	0.848
	LSTM	0.892	0.952	0.949	0.903	0.862	0.859	0.866	0.861
	GRU	0.922	0.968	0.973	0.92	0.876	0.88	0.874	0.884
	CNN-LSTM	0.928	0.972	0.978	0.924	0.896	0.906	0.898	0.906
	TCN	0.958	0.982	0.985	0.957	0.917	0.921	0.918	0.923
35 cm	Proposed	0.964	0.984	0.988	0.95	0.951	0.956	0.943	0.944
	KNN	0.859	0.917	0.921	0.859	0.845	0.838	0.837	0.831
	Decision Tree	0.859	0.925	0.93	0.859	0.846	0.838	0.839	0.839
	SVM	0.866	0.935	0.938	0.872	0.848	0.839	0.842	0.845
	CNN	0.87	0.936	0.938	0.897	0.858	0.856	0.85	0.852
	LSTM	0.892	0.949	0.953	0.892	0.859	0.856	0.867	0.86
	GRU	0.921	0.962	0.965	0.925	0.86	0.862	0.87	0.867
	CNN-LSTM	0.916	0.965	0.968	0.904	0.896	0.893	0.897	0.892
40 cm	TCN	0.963	0.978	0.981	0.952	0.926	0.927	0.928	0.929
	Proposed	0.965	0.988	0.987	0.963	0.946	0.96	0.948	0.955
	KNN	0.852	0.921	0.92	0.855	0.841	0.843	0.835	0.838
	Decision Tree	0.862	0.932	0.935	0.855	0.842	0.839	0.84	0.836
	SVM	0.868	0.937	0.94	0.87	0.85	0.841	0.845	0.847
	CNN	0.864	0.942	0.94	0.898	0.857	0.849	0.858	0.855
	LSTM	0.892	0.951	0.956	0.897	0.851	0.862	0.868	0.867
	GRU	0.916	0.952	0.956	0.905	0.866	0.862	0.861	0.866
45 cm	CNN-LSTM	0.916	0.965	0.968	0.909	0.896	0.897	0.906	0.902
	TCN	0.963	0.983	0.986	0.955	0.926	0.927	0.928	0.929
	Proposed	0.979	0.989	0.99	0.978	0.96	0.958	0.962	0.948
	KNN	0.847	0.914	0.921	0.845	0.841	0.835	0.836	0.828
	Decision Tree	0.857	0.921	0.928	0.855	0.846	0.835	0.837	0.84
	SVM	0.863	0.929	0.931	0.868	0.846	0.84	0.841	0.845
	CNN	0.865	0.922	0.932	0.889	0.854	0.851	0.85	0.853
	LSTM	0.886	0.946	0.948	0.891	0.855	0.857	0.862	0.866
50 cm	GRU	0.931	0.96	0.971	0.93	0.869	0.868	0.874	0.861
	CNN-LSTM	0.92	0.975	0.967	0.916	0.895	0.899	0.894	0.901
	TCN	0.968	0.988	0.984	0.954	0.932	0.929	0.934	0.927
	Proposed	0.984	0.996	0.998	0.986	0.965	0.966	0.959	0.964
	KNN	0.851	0.918	0.923	0.843	0.845	0.839	0.838	0.833
	Decision Tree	0.862	0.925	0.929	0.851	0.848	0.839	0.842	0.851
	SVM	0.865	0.935	0.934	0.869	0.849	0.848	0.849	0.853
	CNN	0.892	0.941	0.938	0.868	0.852	0.854	0.853	0.858
Mixed	LSTM	0.898	0.947	0.958	0.895	0.854	0.859	0.861	0.865
	GRU	0.934	0.968	0.964	0.919	0.864	0.86	0.869	0.862
	CNN-LSTM	0.928	0.968	0.962	0.913	0.884	0.891	0.89	0.893
	TCN	0.952	0.988	0.989	0.959	0.933	0.928	0.93	0.929
	Proposed	0.981	0.994	0.996	0.982	0.951	0.953	0.955	0.958
	KNN	0.842	0.912	0.919	0.839	0.835	0.831	0.834	0.83
	Decision Tree	0.857	0.921	0.928	0.855	0.846	0.835	0.837	0.84
	SVM	0.863	0.929	0.931	0.868	0.846	0.84	0.841	0.845
Mixed	CNN	0.859	0.946	0.938	0.861	0.863	0.859	0.851	0.856
	LSTM	0.868	0.953	0.962	0.892	0.881	0.864	0.867	0.875
	GRU	0.886	0.966	0.968	0.89	0.891	0.892	0.889	0.885
	CNN-LSTM	0.914	0.972	0.971	0.902	0.906	0.908	0.904	0.909
	TCN	0.922	0.991	0.998	0.947	0.936	0.931	0.928	0.938
	Proposed	0.971	0.989	0.986	0.969	0.946	0.962	0.953	0.961

Table 9

The accuracy of different materials for the AE-CS-TCN model of different models at LMT dataset.

Dataset	Model	Mesher	Stone	Glossy	Wood	Rubbers	Fibers	Foams	Foils and Papers	Textiles and Fabrics
LMT	KNN	0.871	0.875	0.867	0.891	0.868	0.889	0.892	0.832	0.829
	Decision Tree	0.880	0.886	0.874	0.891	0.869	0.887	0.893	0.845	0.849
	SVM	0.886	0.893	0.881	0.894	0.876	0.889	0.892	0.851	0.866
	CNN	0.898	0.905	0.896	0.905	0.884	0.896	0.901	0.871	0.876
	LSTM	0.913	0.917	0.896	0.926	0.890	0.911	0.918	0.891	0.896
	GRU	0.902	0.926	0.915	0.942	0.908	0.921	0.935	0.894	0.891
	CNN-LSTM	0.935	0.978	0.932	0.974	0.935	0.962	0.972	0.898	0.894
	TCN	0.962	0.992	0.942	0.965	0.927	0.981	0.991	0.905	0.902
Mixed	Proposed	0.975	0.982	0.952	0.985	0.946	0.986	0.998	0.915	0.913

from raw input data. Traditional machine learning approaches, although computationally efficient, typically depend on handcrafted feature extraction and are inherently constrained in their ability to model intricate, nonlinear relationships present in material surface characteristics. In contrast, deep learning architectures (including CNN, LSTM, and GRU) are capable of capturing hierarchical features and temporal dynamics, thereby offering more effective representations of material textures and spatial configurations. Despite these advantages, such models exhibit certain structural limitations that hinder their performance. CNNs, for instance, are less effective in modeling temporal dependencies, while RNN based architectures such as LSTM and GRU are prone to issues such as vanishing gradients and suffer from limited parallelization. Even CNN-LSTM may fail to adequately model long range temporal correlations in sequential sensor data. Temporal Convolutional Networks demonstrate enhanced temporal modeling capabilities through the use of dilated causal convolutions and residual connections, leading to competitive performance especially under mixed distance conditions. However, the proposed model goes beyond the capabilities of TCN by integrating advanced mechanisms for multi scale feature fusion and adaptive attention to salient spatiotemporal patterns. These architectural enhancements enable the model to effectively encode both localized fine grained features and global contextual information, which is essential for distinguishing between materials that exhibit subtle differences in surface properties, such as foams, textiles, and thin films. Furthermore, the consistently low standard deviation observed across multiple experimental trials, not exceeding 0.008, demonstrates the model's robustness and reliability under varying conditions. This level of consistency is critical for real world deployment, where environmental variability and sensor noise can introduce uncertainty into the input data. The inclusion of spatial attention and multi resolution temporal aggregation within the proposed framework not only enhances its discriminative capability but also contributes to improved model interpretability, thereby facilitating a more rigorous understanding of the underlying decision making process.

6. Conclusion

Experimental results validate the effectiveness of AE-CS-TCN: across datasets collected at varying distances (30–50 cm) and a mixed-distance scenario, the model achieves an average accuracy of 96%, with peak performance (97.7%) at 45 cm. On the LMT public dataset, which includes 9 material categories such as meshes, stone, and textiles, AE-CS-TCN consistently outperforms ablated model variants, achieving the highest accuracy across most categories. Notably, it reaches 0.998 accuracy for foams, 0.985 for wood, and 0.986 for fibers, demonstrating strong generalization to diverse material types beyond the custom dataset.

Comparisons with state-of-the-art methods, including KNN, SVM, CNN, and TCN, confirm the superior discriminative capability of AE-CS-TCN, particularly in distinguishing materials with subtle acoustic differences. Ablation studies further demonstrate that each component contributes to performance, with spatial attention and cross-scale fusion playing key roles. The custom experimental system and validation on the LMT dataset ensure results generalize to real-world robotic applications, while data augmentation via EMD-WGN mitigates overfitting.

Future work will expand the material library and optimize for real-time inference. Integrating multi-modal sensing could further enhance reliability. Overall, AE-CS-TCN advances non-contact material recognition, offering a practical solution for robotic perception in harsh environments, with validated performance across both custom and public datasets.

CRediT authorship contribution statement

Bo Zhu: Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Haoxuan Li:** Software, Resources, Investigation. **Tao Geng:** Supervision, Resources, Investigation, Funding acquisition. **Wenqiang Duan:** Project administration, Methodology, Funding acquisition. **Boxin Ren:** Validation, Software, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.sigpro.2025.110408>.

Data availability

Data will be made available on request.

References

- [1] Geon Kang, Seung-Chan Kim, Deepecho: Echoacoustic recognition of materials using returning echoes with deep neural networks, *IEEE Trans. Emerg. Top. Comput.* 10 (1) (2022) 450–462.
- [2] Xiling Yao, Shumin Liu, Jie Chen, Shefeng Yan, Fei Ji, Hongwei Liu, Jingdong Chen, Underwater acoustic target classification using scattering transform with small sample size, *IEEE Sens. J.* 24 (16) (2024) 25998–26010.
- [3] Wei Wang, Shefeng Yan, Linlin Mao, Zeping Sui, Jirui Yang, Robust direct position determination for chirp signal-based underwater acoustic sensor networks, *Signal Process.* 230 (2025).
- [4] Xu Liu, Junzheng Wang, Jiehao Li, Urtsegnet: A real-time segmentation network of unstructured road at night based on thermal infrared images for autonomous robot system, *Control Eng. Pract.* 137 (2023) 105560.
- [5] Qin Tang, Jing Liang, Fangqi Zhu, A comparative review on multi-modal sensors fusion based on deep learning, *Signal Process.* 213 (2023) 109165.
- [6] Bo Zhu, Tao Geng, Baoyi Wang, Haoxuan Li, Xianhong Zhang, Robust non-contact material recognition for robots in extreme and dynamic environments, *Comput. Industry (ISSN: 0166-3615)* 174 (2026) 104411, <http://dx.doi.org/10.1016/j.compind.2025.104411>.
- [7] Bo Zhu, Tao Geng, Gedong Jiang, Zheng Guan, Yang Li, Xialun Yun, Surrounding object material detection and identification method for robots based on ultrasonic echo signals, *Appl. Bionics Biomech.* 2023 (2023) 1–12.
- [8] Yakun Shi, Wanli Xu, Jun Zhang, Xiaohong Li, Automated classification of ultrasonic signal via a convolutional neural network, *Appl. Sci.* 12 (9) (2022).
- [9] Jorge Wuth, Rodrigo Mahu, Israel Cohen, Richard M. Stern, Néstor Becerra Yoma, A unified beamforming and source separation model for static and dynamic human–robot interaction, *JASA Express Lett.* 4 (3) (2024) Cited by: 2; All Open Access, Gold Open Access.
- [10] Yu Wang, Haoyao Chen, Yunhui Liu, Wenqiang Chen, Eil-slam: Depth-enhanced edge-based infrared-lidar slam, *J. Field Robot.* 39 (2) (2022) 117–130.
- [11] Roberto Longo, S. Vanlanduit, Patrick Guillaume, Material properties identification using ultrasonic waves and laser doppler vibrometer measurements: A multi-input multi-output approach, *Meas. Sci. Technol.* 24 (2013) 105206.
- [12] Sriram Siva, Hao Zhang, Robot perceptual adaptation to environment changes for long-term human teammate following, *Int. J. Robot. Res.* 41 (7) (2022) 706–720.
- [13] Youcan Yan, Zhe Hu, Zhengbao Yang, Wenzhen Yuan, Chaoyang Song, Jia Pan, Yajing Shen, Soft magnetic skin for super-resolution tactile sensing with force self-decoupling, *Sci. Robot.* 6 (51) (2021) 121–135.
- [14] Ancong Wu, Wei Shi Zheng, Shaogang Gong, Jianhuang Lai, Rgb-ir person re-identification by cross-modality similarity preservation, *Int. J. Comput. Vis.* 128 (8) (2020).
- [15] Hongyi Liu, Lihui Wang, Remote human–robot collaboration: A cyber–physical system application for hazard manufacturing environment, *J. Manuf. Syst.* 54 (2020) 24–34.
- [16] Hideto Otsuka, Kan Okubo, Non-contact identification of moisture content of fabric based on analysis of broadband acoustic signals using multiple-frequency air ultrasonic transducer system, *Japan. J. Appl. Phys.* 61 (2022) 154–168.
- [17] Min-Hyun Kim, JongChan Park, Seibum Choi, Road type identification ahead of the tire using d-cnn and reflected ultrasonic signals, *Int. J. Automot. Technol.* 22 (2021) 47–54.

- [18] Biao Wu, Wensong Zhou, Ultrasonic defect detection in noisy signals by a nonconvex sparse regularization approach, *Appl. Acoust.* 210 (2023) 109461.
- [19] Gangfeng Zheng, Wandong Xia, Liangbo Ma, Wei Sun, Chao Sun, Yi'ao Xu, Xiaodong Zhao, Kaifeng Wang, An experimental method to measure the layer thickness and wave velocity of copper-steel composite board without interface echo, *Measurement* 91 (2016) 77–83.
- [20] Aleksandr Bystrov, Mohammad Abbas, Edward Hoare, Thuy-Yung Tran, Nigel Clarke, Marina Gashinova, Mikhail Cherniakov, Remote road surface identification using radar and ultrasonic sensors, in: 2014 11th European Radar Conference, 2014, pp. 185–188.
- [21] Aleksandr Bystrov, Edward Hoare, Thuy-Yung Tran, Nigel Clarke, Marina Gashinova, Mikhail Cherniakov, Road surface classification using automotive ultrasonic sensor, *Procedia Eng.* 168 (2016) 19–22.
- [22] Guangdong Zhang, Xiongbing Li, Shuzeng Zhang, Tribikram Kundu, Investigation of frequency-dependent attenuation coefficients for multiple solids using a reliable pulse-echo ultrasonic measurement technique, *Measurement* 177 (2021) 109270.
- [23] Huifang Xiao, Dan Chen, Jinwu Xu, Shifeng Guo, Defects identification using the improved ultrasonic measurement model and support vector machines, *NDT & E Int.* 111 (2020) 102223.
- [24] Zafiris Politis, Penny Probert Smith, Classification of textured surfaces for robot navigation using continuous transmission frequency-modulated sonar signatures, *Int. J. Robot. Res.* 20 (2001) 107–128.
- [25] Ruth Thomas, Bruce Drinkwater, Dimosthenis Liaptsis, The reflection of ultrasound from partially contacting rough surfaces, *J. Acoust. Soc. Am.* 117 (2005) 638–645.
- [26] Mohsen Mousavi, Mohammad Sadegh Taskhiri, Damien Holloway, J.C. Olivier, Paul Turner, Feature extraction of wood-hole defects using empirical mode decomposition of ultrasonic signals, *NDT & E Int.* 114 (2020) 102282.
- [27] Roberto Longo, Quentin Grimal, Pascal Laugier, S. Vanlanduit, Patrick Guillaume, Simultaneous determination of acoustic velocity and density of a cortical bone slab: Ultrasonic model-based approach, *Ultrason. Ferroelectr. Freq. Control. IEEE Trans.* 57 (2010) 496–500.
- [28] Yehav Alkaher, Israel Cohen, Howling detection and gain control for speech reinforcement in a noisy car cabin environment, *IEEE/ACM Trans. Audio Speech Lang. Process.* 32 (2024) 1494–1505, Cited by: 1.
- [29] Longbin Yan, Min Zhao, Shumin Liu, Shuaikai Shi, Jie Chen, Cascaded transformer u-net for image restoration, *Signal Process.* 206 (2023).
- [30] Eisele Jona, Gerlach André, Maeder Marcus, Marburg Steffen, Convolutional neural network with data augmentation for object classification in automotive ultrasonic sensing, *J. Acoust. Soc. Am.* 153 (4) (2023) 2447–2459.
- [31] Penny Probert Smith, Konstantinos Zografos, Sonar for recognising the texture of pathways, *Robot. Auton. Syst.* 51 (1) (2005) 17–28.
- [32] Hideto Otsuka, Kan Okubo, Non-contact identification of moisture content of fabric based on analysis of broadband acoustic signals using multiple-frequency air ultrasonic transducer system, *Japan. J. Appl. Phys.* 61 (2022).
- [33] Yuchen Dong, Jie Chen, Wen Zhang, Wave-domain active noise control over distributed networks of multi-channel nodes *, *Signal Process.* 184 (2021).
- [34] Vahid Nasir, Yousef Parvari, Hamidreza Fathi, Siavash Kazemirad, Farrokh Sassani, Identification of wood species and duration of exposure in weathered wood using guided wave propagation, *Wood Mater. Sci. Eng.* 18 (2022) 1–12.
- [35] Hongyi Liu, Lihui Wang, Human motion prediction for human-robot collaboration, *J. Manuf. Syst.* 44 (2017) 287–294.
- [36] Ajit Kumar Sahoo, Siba K. Udgata, Material classification based on non-contact ultrasonic echo signal using deep learning approach, *Procedia Comput. Sci.* 235 (2024) 606–616, International Conference on Machine Learning and Data Engineering (ICMLDE 2023).
- [37] Mayuka Kono, Yutaro Hirao, Monica Perusquía-Hernández, Naoya Isoyama, Hideaki Uchiyama, Nobuchika Sakata, Jun Takamatsu, Kiyoshi Kiyokawa, U2r: Underwater ultrasonic reflection wave dataset toward pose-invariant material recognition, in: ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, 2024, pp. 646–650.
- [38] Bing Yan, Peng Shi, Chee Peng Lim, Yuan Sun, Ramesh K. Agarwal, Security and safety-critical learning-based collaborative control for multiagent systems, *IEEE Trans. Neural Networks Learn. Syst.* 36 (2) (2025) 2777–2788.
- [39] Markku Tiitta, V. Tiitta, Mate Gaal, J. Heikkinen, Reijo Lappalainen, Laura Tomppo, Air-coupled ultrasound detection of natural defects in wood using ferroelectret and piezoelectric sensors, *Wood Sci. Technol.* 54 (2020) 1–14.
- [40] Juan J. González, Jovani A. Jiménez, Jaime A. Guzmán, Peniel method for the ultrasonic material identification in robots, in: 2011 IEEE International Symposium on Signal Processing and Information Technology (ISSPIT), vol. 45, 2011, pp. 496–501.
- [41] Jürgen Schmidhuber, Deep learning in neural networks: An overview, *Neural Netw.* 61 (2015) 85–117.
- [42] Hassan Ismail Fawaz, Germain Forestier, Jonathan Weber, Lhassane Idoumghar, Pierre-Alain Muller, Deep learning for time series classification: a review, *Data Min. Knowl. Discov.* 33 (2019) 16–25.
- [43] Thorsten Hoeser, Felix Bachofer, Claudia Kuenzer, Object detection and image segmentation with deep learning on earth observation data: A review-part ii: Applications, *Remote. Sens.* 12 (2020) 3053.
- [44] Jona Eisele, André Gerlach, Marcus Maeder, Steffen Marburg, Relevance of phase information for object classification in automotive ultrasonic sensing using convolutional neural networks, *J. Acoust. Soc. Am.* 155 (2024) 1060–1070.
- [45] Qichun Hu, Xiaolong Wei, Hanyi Guo, Haojun Xu, Caizhi Li, Weifeng He, Binbin Pei, Study on intelligent and visualization method of ultrasonic testing of composite materials based on deep learning, *Appl. Acoust.* 207 (2023) 109363.
- [46] Xuewen Qin, Feng Gao, Junyu Dong, Xinbo Gao, Sawu-net: Spatial attention weighted unmixing network for hyperspectral images, *IEEE Geosci. Remote. Sens. Lett.* PP (2023) 1–16.
- [47] Amy Ni, John Maunsell, Neuronal effects of spatial and feature attention differ due to normalization, *J. Neurosci.* 39 (2019) 06–18.
- [48] Xian-Feng Han, Zhang-Yue He, Jia Chen, Guoqiang Xiao, 3Crossnet: Cross-level cross-scale cross-attention network for point cloud representation, *IEEE Robot. Autom. Lett.* 7 (2022).
- [49] He Li, Junge Zhang, Weihang Kong, Jienan Shen, Yuguang Shao, Csa-net: Cross-modal scale-aware attention-aggregated network for rgb-t crowd counting, *Expert Syst. Appl.* 213 (2022) 119038.
- [50] Zong Meng, Wei Cao, Dengyun Sun, Qian Li, Wuxu Ma, Fengjie Fan, Research on fault diagnosis method of ms-cnn rolling bearing based on local central moment discrepancy, *Adv. Eng. Informatics* 54 (2022) 101797.
- [51] Shanghua Gao, Ming-Ming Cheng, Kai Zhao, Xin-Yu Zhang, Ming-Hsuan Yang, Philip Torr, Res2net: A new multi-scale backbone architecture, *IEEE Trans. Pattern Anal. Mach. Intell.* PP (2019) 1–1.
- [52] Matti Strese, Rania Hassen, Andreas Noll, Eckehard Steinbach, A tactile computer mouse for the display of surface material properties, *IEEE Trans. Haptics* 12 (1) (2019) 18–33.
- [53] Matti Strese, Clemens Schuwerk, Albert Iepure, Eckehard Steinbach, Multimodal feature-based surface material classification, *IEEE Trans. Haptics* 10 (2) (2017) 226–239.
- [54] Tonya Fields, George Hsieh, Jules Chenou, Mitigating drift in time series data with noise augmentation, in: 2019 International Conference on Computational Science and Computational Intelligence, CSCI, 2019, pp. 227–230.
- [55] Terry T. Um, Franz M.J. Pfister, Daniel Pichler, Satoshi Endo, Muriel Lang, Sandra Hirche, Urban Fietzek, Dana Kulić, Data augmentation of wearable sensor data for parkinson's disease monitoring using convolutional neural networks, in: Proceedings of the 19th ACM International Conference on Multimodal Interaction, ICMi 2017, ACM, New York, NY, USA, 2017, pp. 216–220.
- [56] Justin Salamon, Juan Bello, Deep convolutional neural networks and data augmentation for environmental sound classification, *IEEE Signal Process. Lett.* PP (2017).
- [57] Cédric Rommel, Joseph Paillard, Thomas Moreau, Alexandre Gramfort, Data augmentation for learning predictive models on eeg: a systematic comparison, *J. Neural Eng.* 19 (2022).
- [58] Aswathy Madhu, Suresh Kumaraswamy, Envgan: a gan-based augmentation to improve environmental sound classification, *Artif. Intell. Rev.* 55 (2022) 1–20.
- [59] Matthew C. Cieslak, Ann M. Castelfranco, Vittoria Roncalli, Petra H. Lenz, Daniel K. Hartline, T-distributed stochastic neighbor embedding (t-sne): A tool for eco-physiological transcriptomic analysis, *Mar. Genom.* 51 (2020) 100723.
- [60] Sharnil Pandya, Hemant Ghayvat, Ambient acoustic event assistive framework for identification, detection, and recognition of unknown acoustic events of a residence, *Adv. Eng. Informatics* 47 (2021) 101238.
- [61] Honghui Yang, Kaifeng Zheng, Li Junhao, Open set recognition of underwater acoustic targets based on gru-cae collaborative deep learning network, *Appl. Acoust.* 193 (2022) 108774.
- [62] Muhammad Akmal, Tensor factorization and attention-based cnn-lstm deep-learning architecture for improved classification of missing physiological sensors data, *IEEE Sensors J.* 23 (2) (2023) 1286–1294.
- [63] Panagiotis Tsinganos, Bart Jansen, Jan Cornelis, Athanasios Skodras, Real-time analysis of hand gesture recognition with temporal convolutional networks, *Sensors* 22 (2022) 1694.